

Local Area Augmentation of GPS for the Precision Approach of Aircraft

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The Federal Aviation Administration (FAA) is investigating the feasibility of using the Global Positioning System (GPS), together with local augmentation, as the primary navigation system for the Category II and III precision approach of aircraft. This augmented system would be known as a local area augmentation system (LAAS) because it would place subsystems on the ground at or near an airport to greatly enhance the performance of airborne GPS receivers approaching that airport. The LAAS would complement space-based augmentations of GPS, such as the Wide Area Augmentation System.

When complete, the integrated system will meet stringent requirements on accuracy, integrity, continuity, and availability. Indeed, the LAAS will improve the airborne accuracy from approximately 100 m for stand-alone GPS to better than 1 m. This high accuracy is required so that LAAS can guarantee the integrity and continuity of the aircraft guidance during the approach phase of flight. Integrity requires the LAAS to detect any situations which threaten the safety of the landing, and notify the aircraft within 2 s of such a threat. Continuity is a competing consideration—it requires that the total number of alarms, true plus false, sent to the aircraft is very small. High continuity guarantees that the total number of aborted approaches is manageably small. Finally, availability is the fraction of the time during which LAAS is operational—providing position fixes with the specified accuracy, integrity and continuity.

This paper describes the LAAS which has been proposed by the FAA. It discusses the fundamental operation of the LAAS, and focuses on the ranging signals used to estimate the aircraft position. It does not provide the details of the integrity monitoring algorithms or of the ground-to-air datalink. These topics are well described elsewhere in the maturing literature on these subjects. However, this paper does establish the connection between accuracy, integrity and continuity. It also discusses how the required accuracy will be provided with adequate availability. It also lists the issues which are currently being resolved.

Keywords—Aircraft landing, Global Positioning System, precision approach, radio navigation, satellite navigation.

I. INTRODUCTION

In time, Global Positioning System (GPS) will be used for a wide variety of aircraft operations. These will include flight over oceanic routes and en route through our domestic airspace. GPS will also be used in the so-called termi-

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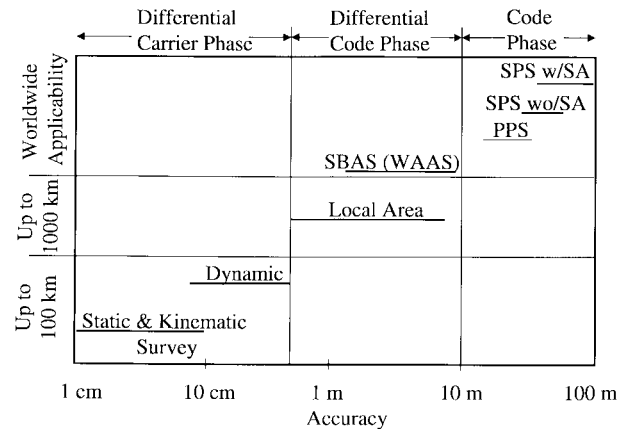


Fig. 1. Hierarchy of GPS capabilities.

nal areas where flights converge in crowded metropolitan airspaces. Examples of heavily used terminal areas include the New York–Newark and the San Francisco Bay areas with their multiplicity of airports. GPS will be used on the final approach to airports—these operations demand the greatest safety and reliability. Airport approach applications include: nonprecision approach, where GPS will be used solely for horizontal positioning, and precision approach where GPS will be used for both horizontal and vertical position fixing. To serve all these applications, GPS must be augmented to meet stringent requirements with respect to accuracy, integrity, continuity, and time availability.

The introduction to this paper will describe the aviation applications served by the hierarchy of GPS capabilities shown in Fig. 1. The body of the paper will concentrate on the use of local area augmentation systems to support precision approach operations. However, we must first introduce the language which is used to describe all aviation navigation systems.

A. Definitions

Accuracy is the difference between the estimated and true aircraft position. More specifically, *navigation sensor error* (NSE) is defined as the difference between the estimated and true position. NSE is the error which is only exceeded 5% of the time in the absence of system failures.

Integrity and *continuity* characterize the system response to component failures or rare natural events—they are the main measures of flight safety. They address the ability of a navigation system to detect and possibly repair system threats in a timely fashion.

Integrity fails when the position error exceeds a certain *alert limit* (AL, measured in meters), and this error is not annunciated to the pilot or the aircraft guidance system within a specified *time-to-alarm*. *Integrity risk* is defined as the probability that all alerts are silent while the aircraft's position error exceeds the specified AL for a time greater than the time-to-alarm. Precision approach operations have specifications for the horizontal alert limit (HAL), vertical alert limit (VAL), and time-to-alarm.

In contrast to integrity, continuity fails when an aircraft operation must be aborted for any unscheduled reason. *Continuity risk* is the probability that the navigation system will fail during a flight operation given that it was available at the beginning of the operation. Continuity and integrity are competing requirements, because continuity requires that the total number of missed approaches is very small. Consequently, the integrity algorithms cannot be overly sensitive. If they are unduly sensitive, too many false alarms will be sent to the aircraft and unnecessary missed approaches will be conducted.

Time availability is the fraction of time for which the system is operational—providing position fixes with the specified accuracy, integrity, and continuity. In contrast to integrity and continuity, availability is not an instantaneous measure of flight safety. Rather it emphasizes the operational acceptability or economy of a navigation system.

We now turn our attention to categorizing air navigation systems based on their ability to provide accuracy, continuity, integrity, and availability [1].

A *sole-means* navigation system for a certain flight mode (e.g., nonprecision approach) meets all the requirements (integrity, continuity, accuracy, and availability) for that mode of flight. If a sole-means system is onboard, then no other system is required for that flight mode. A *primary-means* navigation system for a certain flight mode need only meet the integrity and accuracy requirements for that operation. The system need not meet the availability and continuity requirements. Nonetheless, a sole-means system need not accompany a primary means system because suitable procedural restrictions and backup procedures have been defined. For example, flight departure times may be restricted to avoid times when the system is known to be unavailable or known to have high continuity risk. In the case of a GPS-based system, departures could be prohibited during periods when too few satellites would be available to complete the operation.

A *supplemental* navigation system also meets the integrity and accuracy requirements for a given mode of flight. Like a primary-means system, it need not meet the requirements for continuity and/or availability. In this case however, a sole-means system is carried to compensate for the outages.

B. Standard and Precise Positioning Services

The Standard Positioning Service (SPS) provides civilian users with guaranteed access to a 1-MHz spreading code (the coarse/acquisition (C/A) code), which modulates a signal at $f_{L1} = 1575.42$ MHz [27]. With Selective Availability (SA) enabled, the SPS provides 100-m horizontal accuracy (95%). If SA were to be turned off, then the accuracy of SPS would be around 10–20 m. The Precise Positioning Service (PPS) provides users authorized by the U.S. Department of Defense with access to a 10-MHz code (the P or Y code), which modulates the signal at $f_{L1} = 1575.42$ MHz and a signal at $f_{L2} = 1227.60$ MHz. The secure modulation of f_{L1} is in quadrature to the C/A code modulation of f_{L1} . The PPS provides a horizontal accuracy of approximately 10 m (95%). These capabilities of the SPS and PPS are shown in Fig. 1.

The SPS and the PPS meet the accuracy requirements for en-route flight, terminal area operations, and nonprecision approach. Of these operations, nonprecision approach has the most demanding accuracy requirement—100 m in the horizontal. However, neither the SPS nor the PPS are capable of meeting any of the other requirements—integrity, continuity or availability—without augmentation of some sort.

Satellite navigation differs markedly from land-based equipment with respect to the nature of availability. Land-based equipment provides an accuracy which does not vary with time—until a critical piece of hardware malfunctions. When such a failure occurs, the outage is usually conspicuous. In contrast, GPS outages need not be caused by the failure of a piece of hardware. For example, accuracy may decrease simply because one satellite has disappeared over the user's horizon, and this degradation may persist until another satellite rises. The setting and rising of GPS satellites is completely predictable. Nonetheless, SPS and PPS outages due to too few satellites move across the airspace and may make certain flight operations in certain areas unavailable for tens of minutes.

To illustrate this variability, the top traces in Figs. 2 and 3 show SPS performance over 24 h for a user at San Francisco International Airport. They show the approximate vertical accuracy provided by the SPS with SA. (Horizontal accuracy is approximately 50% better than vertical accuracy.) Fig. 2 shows the variation of accuracy when all 24 satellites in the GPS constellation are healthy and transmitting their signals. Fig. 3 shows the accuracy when only 21 of 24 satellites are active. Clearly, the sparser constellation suffers a greater variation in performance. As described later in this paper, this sensitivity can be ameliorated by placing additional ranging sources on geostationary satellites or on the ground using pseudosatellites (pseudolites).

SPS or PPS use for aviation raises integrity as a concern—especially since a single undetected satellite fault could effect many aircraft over continental areas. The GPS ground control segment (GCS) certainly provides a degree of integrity. It monitors all of the GPS satellites, and updates the broadcast health message if any problems

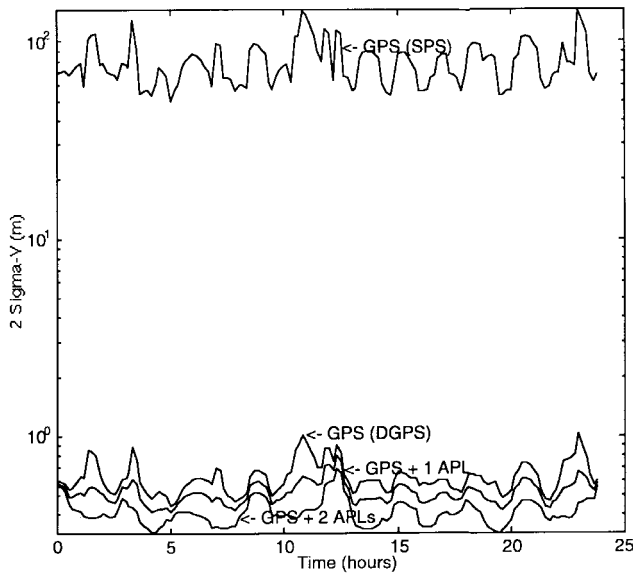


Fig. 2. Vertical accuracy (2 sigma) versus time for an aircraft approaching San Francisco International Airport when 24 GPS satellites are operational ($C_{\text{zenith}}/N_o = 40$ dB-Hz, $T_{\text{integration}} = 25$ s, elevation mask is 5°).

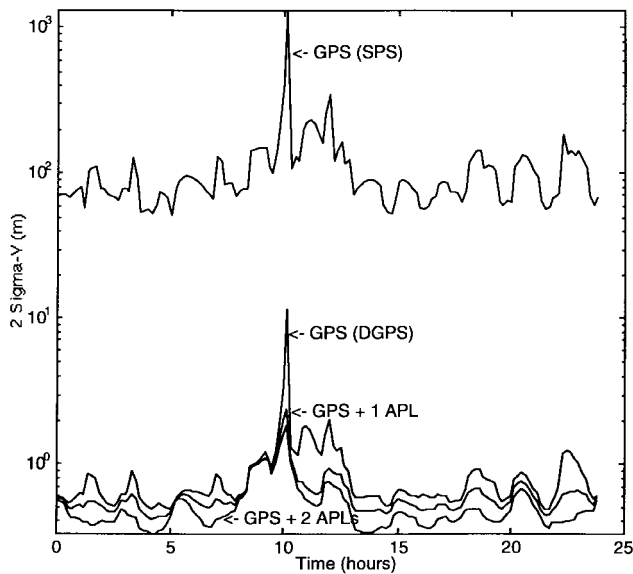


Fig. 3. Vertical accuracy (2 sigma) versus time for an aircraft approaching San Francisco International Airport when 21 GPS satellites are operational ($C_{\text{zenith}}/N_o = 40$ dB-Hz, $T_{\text{integration}} = 25$ s, elevation mask is 5°).

are evident. However, such warnings are not necessarily quick enough to protect the safety of an approaching aircraft. Moreover, the GCS does not monitor all GPS satellites continuously, and human error has resulted in the transmission of faulty data. As a result, GPS satellites have suffered from problems that were not immediately flagged by the navigation message. For example, the onboard clocks have exhibited drifts that were not well described by the clock information in the broadcast health message.

Accordingly, autonomous fault detection within the GPS receiver is of great interest, and today's GPS receivers use a wide variety of internal reasonability checks. In particular,

receiver autonomous integrity monitoring (RAIM) refers to self-contained algorithms that test the suite of satellite measurements for self-consistency, often by examining the measurement residuals. In other words, they compare the measured pseudoranges to predicted pseudoranges based on the estimated position of the aircraft. A fault is declared if there is an unexpected disagreement between one (or more) of the actual measurements and the predicted measurements. As such, RAIM is sensitive to discord or inconsistency in the measurements, and it can detect errors which have not yet been flagged by the GCS. For these reasons, GPS receivers with RAIM can be certified for supplemental aircraft use under Technical Standard Order C-129 [2], [3]. These receivers can only be used for en-route flight, terminal area operations, and nonprecision approach—they are not to be used for precision approach.

As mentioned, a TSO-C 129 receiver can only supplement another sole-means navigation system. By itself, a TSO-C 129 receiver is not a sole means or primary means aid to navigation, because the time availability of RAIM is not adequate. If SA is on, then the availability of RAIM for nonprecision approach and terminal area navigation is 94% and 99%, respectively—provided all 24 GPS satellite are operating [4]. If only 21 satellites are operating, then availability falls to 74% and 93%, respectively. RAIM can also be used to identify which satellite is faulty, but this isolation function has a smaller availability than the detection function.

The availability of aircraft autonomous integrity monitoring can be increased by using other position sensors onboard the aircraft. For example, the use of information from the baro-altimeter is considered in [5] and the use of Loran-C to aid GPS is described in [6]. The use of the Russian satellite navigation system GLONASS is well described by Daly and Misra [7]. Finally, the integration of GPS and inertial sensors is analyzed by Brenner [8], Diesel [9], and Greenspan [10]. However, we now turn our attention to space-based augmentation systems (SBAS's) and local-area augmentation systems (LAAS's). These systems improve availability by radiating GPS-like ranging signals from geostationary satellites or ground-based pseudolites.

C. Space-Based Augmentation Systems

SBAS's are being implemented around the world to augment GPS, with major efforts in the United States, Europe, and Japan. As shown in Fig. 4, these systems will deliver [11], [12]:

- ranging signals to improve availability and continuity;
- integrity data to improve flight safety;
- differential corrections to improve accuracy and provide a precision approach capability.

These functions are now briefly described.

To improve availability and continuity, each SBAS will broadcast spread-spectrum ranging signals from geostationary satellites. Since the SBAS signals will be synchronized to GPS, the airborne SBAS receiver will be able to add these new ranging signals to the SPS constellation of

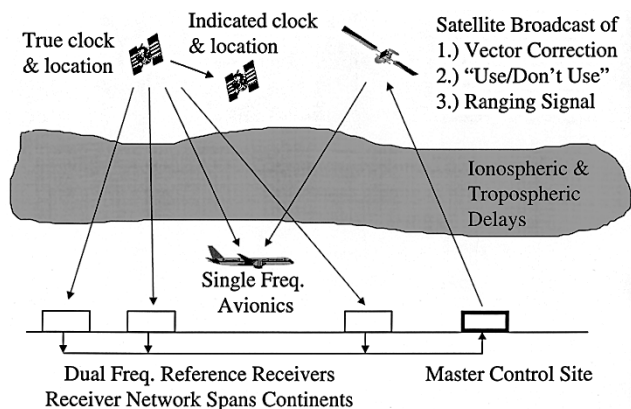


Fig. 4. SBAS.

measurements. This augmented position fix will be less sensitive to the failure of GPS satellites, thus improving time availability and continuity.

To improve integrity, each SBAS will use a network of ground receivers to monitor the health of all satellites over its airspace. This network will span the geographical area of interest with receivers located at the corners of the coverage area. It will detect any satellites which have problems and should not be used for navigation. Appropriate warnings will be modulated on to the SBAS ranging signals and broadcast to the user, thereby improving the integrity of the airborne position fix.

These first two functions will improve dramatically the availability, continuity, and integrity of the SPS. With these improvements alone, SPS plus SBAS will be capable of providing primary means navigation for en-route flight down through nonprecision approach. The third function (differential corrections) is aimed at improving the accuracy of SPS, and by so doing to provide a precision approach capability.

Unlike nonprecision approach, precision approach is based on a constant rate of descent resulting in a smooth glide path. This glide path typically makes an angle of 3° with the horizontal and passes through a decision height (DH) at which the pilot must decide whether or not to complete the landing. Precision approach is divided into three categories that correspond to different weather conditions. The three categories are distinguished from each other by the DH and the runway visual range (RVR), where RVR quantifies visibility along the runway. Category I conditions exist when the decision height is at least 200 ft and the RVR is at least 2400 ft. Category II conditions exist when the decision height is between 100 and 200 ft and the RVR is at least 1200 ft. Category III conditions exist when the decision height is less than 100 ft or the RVR is less than 1200 ft. Category III includes conditions where the aircraft is automatically landed by the flight guidance system. Of course, the requirements become more demanding as the decision height decreases with the most stringent requirements for Category III.

To provide a precision approach capability, each SBAS will use the ground network to develop differential cor-

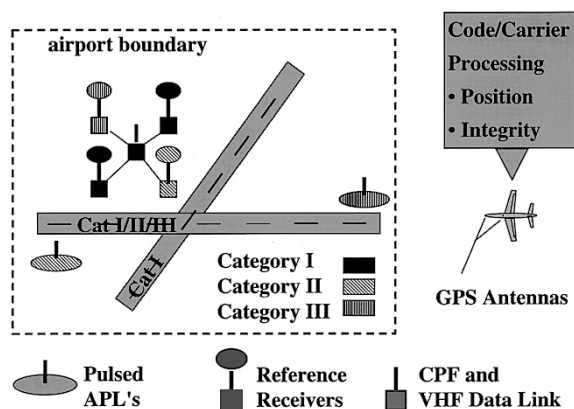


Fig. 5. LAAS.

rections for the errors which limit the accuracy of SPS. The ground network includes a geographically distributed set of GPS receivers at precisely known locations. To enable precision approach, this network must have receivers throughout the coverage area with spacings of approximately 600 km. For example, the SBAS for the United States will place approximately 20 receivers within the coterminous United States. As shown in Fig. 4, these reference stations send raw GPS measurements back to master control stations.

The master stations use the reference observations to develop two corrections for each monitored satellite. One correction is for the satellite clock and the other is a three-dimensional correction for the satellite location. The master station also estimates a set of corrections for the ionospheric delays. All of these corrections will be included on the SBAS broadcast and will improve SPS position accuracy from 100 m to approximately 4 m. With this vertical accuracy, SBAS will provide vertical guidance down to decision heights of 200–350 ft. Lower decision heights require the use of a local area augmentation system.

D. Local-Area Augmentation Systems

LAAS's are being developed by the Federal Aviation Administration (FAA) and the aviation industry to serve all three categories of precision approach. As shown in Fig. 5, an LAAS is a local area differential GPS system. It places a small collection of high-quality GPS reference receivers and antennas at known, surveyed locations on the airport property. These reference measurements (or corresponding corrections) are broadcast to approaching aircraft using a VHF data link. The airborne GPS receiver uses these measurements to correct its own, thereby achieving submeter accuracy. To help insure flight safety, the broadcast also carries data that estimates the quality of the correction and warns of any system malfunctions.

LAAS is a local area differential GPS system because it places all of its reference receivers close together (within tens or hundreds of meters) and only forms a single correction for each satellite. In contrast, SBAS distributes its reference receivers across continental areas and forms multiple corrections for each satellite. In other words,

Table 1

Tentative Requirements for Precision Approach Using LAAS

	CAT I	CAT II	CAT III
Decision Height (DH)	200+ ft	100+ ft	0-100 ft
Vertical Accuracy (V_{nse} , 95%) Required to Control the Aircraft Under Fault-free Conditions	5 m	2.5 m	2.5 m
Continuity	$10^{-5}/\text{app}$	$10^{-5}/\text{app}$	$10^{-7}/30 \text{ s}$
Integrity	$4 \times 10^{-8}/\text{app}$	$4 \times 10^{-8}/\text{app}$	$10^{-9}/\text{app}$
Vertical Alert Limit (VAL)	10 m	5 m	5 m
Time to Alarm	6 sec	2 sec	2 sec
Vertical Accuracy (V_{nse} , 95%) Required for Integrity and Continuity (an approximation from Appendix A)	2 m	1 m	1 m

LAAS forms scalar corrections whereas SBAS forms vector corrections. Vector corrections are better suited for providing wide area coverage because a scalar cannot capture both the local error and its spatial gradients. However, scalar corrections are better for highly demanding applications in local areas. They capture all of the error components which are important close to the reference station, while some local errors cannot be observed by the widely separated SBAS reference stations. For this reason, LAAS is more accurate than SBAS provided the user is within 100 km or so of the reference receiver. Interestingly, LAAS, like SBAS, uses a network of ground receivers, but not to develop vector corrections. Rather, they are used to provide a sensitive and specific check on the integrity of the broadcast corrections.

For Category I and II, LAAS will provide both the vertical and horizontal information down to the decision height. Below the decision height, the aircraft will be controlled manually by the pilot. For a completely automatic landing, LAAS will provide horizontal information to the aircraft as it flares, touches down, and rolls down the runway. However, LAAS will not provide the vertical information to control the final flare maneuver of the landing airplane. Once the aircraft is over the runway, onboard radar altimeters provide the vertical information used to control the aircraft through touchdown. LAAS may provide the horizontal information required for taxi guidance from the runway to the gate.

LAAS may make use of inertial reference systems. Inertial reference systems could provide enough information to replace a satellite signal that failed after the landing operation started. As such, they would have a role in ensuring the continuity of LAAS-based landings.

LAAS will probably include ground-based pseudolites. These pseudolites will radiate GPS-like, spread-spectrum signals from the ground, and the airborne LAAS receiver will add these new ranging signals to the GPS constellation of measurements. Like the SBAS signals from the geostationary satellites, these pseudolite signals augment the aircraft's position fix. With the additional signals, the position solution is less sensitive to the failure of any GPS satellite signals, thus improving time availability and continuity of service.

LAAS will be designed to meet requirements similar to the tentative requirements shown in Table 1. As shown, the vertical accuracy required to control the aircraft under

fault-free conditions is 5 m for Category I and 2.5 m for Categories II and III. The continuity requirements also become more stringent as the decision height decreases and will require continued operation in the presence of most system component failures. For example, the number of satellites in view at the beginning of the approach must be greater than the minimum number required for the operation—in this way a satellite failure will not cause the operation to be aborted.

As shown in the table, the integrity requirement is also demanding. For Category I, the probability that a vertical error greater than 10 m goes undetected for more than 6 s must be less than 4×10^{-8} per approach. For Category III, the vertical alert limit, time-to-alarm, and integrity probability have decreased to 5 m, 2 s, and 10^{-9} , respectively. To meet these requirements, LAAS will use a combination of ground monitoring, reasonability checks inside the airborne receiver, and comparisons between duplicated or triplicated strings of LAAS avionics.

Ground monitoring will compare GPS measurements from two or more of the reference receivers. This comparison is used to detect faulty measurements before they influence the broadcast correction. Such a fault detection algorithm is analyzed in Appendix A to approximately quantify the relationship between accuracy, integrity, and continuity. As shown in Table 1, the Appendix finds that the accuracy required for fault detection is 2.5 times more stringent than the accuracy required to control the aircraft under fault-free conditions.

As shown in Fig. 1, the derived accuracy requirement for Categories II and III (1.2 m) falls close to the boundary between the levels of service provided by local area code phase differential GPS and local area carrier phase differential GPS. As shown in Fig. 6, the GPS code phase measurement is based on measuring the arrival time of a C/A code chip, which has a duration of 1 μs or approximately 300 m. In contrast, carrier phase measurement are based on the wavelength of the GPS signal, which is 19 cm. Users of differential carrier phase GPS measure the phase of the GPS carrier relative to the carrier phase at the reference site, and achieve range measurement precisions which are a few percent of the carrier wavelength. However, this centimeter precision may not provide centimeter accuracy. Indeed, the receiver only provides the fractional carrier phase measurement, and the number of whole wavelengths in the measurement is ambiguous.

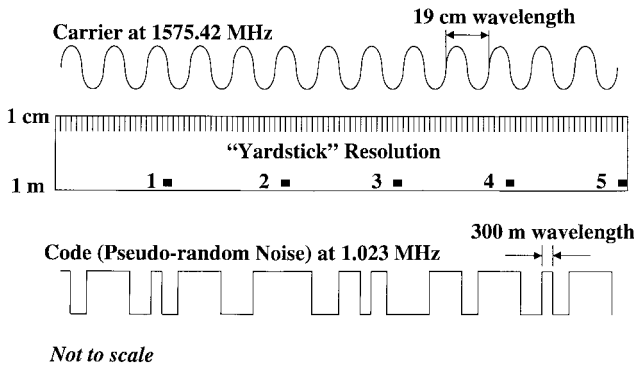


Fig. 6. GPS code and carrier measurements.

The major LAAS dilemma has presented itself. The accuracy of the code phase measurement may not be adequate for ensuring the integrity and continuity of Category II and III landings. The carrier phase measurement certainly has adequate precision, but conversion of this precision to accuracy requires resolution of the whole wavelength ambiguity. As described in the remainder of this paper, LAAS resolves this dilemma in a very elegant fashion.

E. Outline of the Paper

This paper details the multiplicity and accuracy of the measurements used by LAAS. Section II of this report provides a basic introduction to the code and carrier measurements used by GPS. It introduces differential code phase and differential carrier phase positioning. It also discusses carrier smoothed code, where the carrier measurements are used to estimate the change in the aircraft position. Section III analyzes the accuracy of carrier-smoothed measurements when they are used in the aircraft landing environment. Section IV describes the use of airport pseudolites (APL's) including ranging APL's and hyperbolic APL's. Section V is a summary and lists the issues which are currently being considered as part of the architecture validation.

Appendix A considers a specific integrity threat and quantifies the relationship between accuracy, integrity, and continuity. Appendixes B and C bound the multipath Doppler shift and the magnitude of code phase multipath, respectively.

This paper does not detail the data format or VHF link which carries the LAAS data from ground to air. For this information, the reader is referred to the current industry-government effort to standardize this interface [13]. Nor does it describe all of the algorithms and strategies used to trap integrity threats and ensure continuity. This topic is beyond the scope of a single paper and the reader is referred to [14]–[20]. Rather, this paper focuses on the LAAS signals and measurements; these measurements must be accurate. Otherwise, a desirable tradeoff between integrity and continuity cannot be obtained with adequate availability.

II. GPS MEASUREMENTS AND POSITION SOLUTIONS

The measured code and carrier observations at the aircraft and a ground reference receiver are well modeled by

$$\begin{aligned}
 \tau_m^{(k)} &= \left(r_m^{(k)} + b_m - B^{(k)} \right) + I_m^{(k)} + T_m^{(k)} + \nu_{m,\tau}^{(k)} \\
 \tau_u^{(k)} &= \left(r_u^{(k)} + b_u - B^{(k)} \right) + I_u^{(k)} + T_u^{(k)} + \nu_{u,\tau}^{(k)} \\
 \Delta\tau_{m,u}^{(k)} &= \tau_m^{(k)} - \tau_u^{(k)} \approx r_{m,u} \cdot \mathbf{1}^{(k)} + b_{m,u} + \Delta\nu_{m,u,\tau}^{(k)} \\
 \phi_m^{(k)} &= \left(r_m^{(k)} + b_m - B^{(k)} \right) \\
 &\quad - I_m^{(k)} + T_m^{(k)} + N_m^{(k)}\lambda + \nu_{m,\phi}^{(k)} \\
 \phi_u^{(k)} &= \left(r_u^{(k)} + b_u - B^{(k)} \right) - I_u^{(k)} \\
 &\quad + T_u^{(k)} + N_u^{(k)}\lambda + \nu_{u,\phi}^{(k)} \\
 \Delta\phi_{m,u}^{(k)} &= \phi_m^{(k)} - \phi_u^{(k)} \approx r_{m,u} \cdot \mathbf{1}^{(k)} + b_{m,u} \\
 &\quad + \Delta N_{m,u}^{(k)}\lambda + \Delta\nu_{m,u,\phi}^{(k)}.
 \end{aligned} \tag{1}$$

In these equations, the subscripts u and m denote measurements made by the airborne user and the m th reference station, respectively, and the superscript k denotes the k th satellite. For example, $\tau_m^{(k)}$ is the code phase measurement made by the m th reference station for the k th satellite. $\phi_m^{(k)}$ and $\phi_u^{(k)}$ are the carrier phase measurements, converted to units of length, made by the m th reference station and the user, respectively. The code and carrier phase observations share the same pseudorange $\rho^{(k)} = r^{(k)} + b - B^{(k)}$, where $r^{(k)}$ is the true geometric range from the aircraft to the user or reference station. The user clock offset is b_u and the satellite clock offset is $B^{(k)}$. The satellite clock offset includes the effects of SA, and the delays due to the ionosphere and troposphere are denoted $I^{(k)}$ and $T^{(k)}$, respectively. The measurement differences $(\Delta\tau_{m,u}^{(k)}, \Delta\phi_{m,u}^{(k)})$ are discussed in Section II-A.

As shown in (1), the code and carrier measurements both suffer from random observation noise, which is denoted $\nu_{\tau}^{(k)}$ for the code phase and $\nu_{\phi}^{(k)}$ for the carrier. These random variables model the impact of thermal noise, multiple access interference, and multipath. In the absence of multipath, the standard deviation of the carrier noise is approximately one centimeter. As such, the carrier measurement is much more precise than the code measurement, which has noise of approximately 1 m. However, the carrier phase measurement does suffer from a whole wavelength ambiguity $N^{(k)}\lambda$ due to an unknown number of full cycles between the user and the satellite.

As shown in Fig. 7, multipath arises when GPS signals travel over multiple paths from the satellite to the receiver. Some of the signals are delayed relative to the direct signal because they have traveled paths that include a reflection. The reflecting object may be a building, another ship, aircraft or truck, or it may be the surface of the sea or a runway. In general, the more worrisome reflections occur close to the receiver. The figure depicts a reflection from below the ground upon which the reference antenna is placed, because this situation is of particular concern to LAAS.

As mentioned earlier, the measurement differences do still contain errors due to noise, interference, or multipath because these do not cancel. Sadly, these effects can be very large in code phase systems that make no use of the carrier phase.

C. Carrier-Aiding of Code Measurements

Carrier aiding differences the carrier measurements in time and only uses the change in the carrier from one epoch to the next ($\Delta\phi_{m,u}^{(k)}(t_j) - \Delta\phi_{m,u}^{(k)}(t_{j-1})$) to update the current estimate of the pseudorange or position [24]. Carrier aiding can be implemented with the following recursion:

$$\Delta\hat{\rho}_{m,u}^{(k)}(t_j) = \frac{1}{L}\Delta\tau_{m,u}^{(k)}(t_j) + \frac{L-1}{L} \cdot \left[\Delta\hat{\rho}_{m,u}^{(k)}(t_{j-1}) + \Delta\phi_{m,u}^{(k)}(t_j) - \Delta\phi_{m,u}^{(k)}(t_{j-1}) \right] \quad (5)$$

where $\Delta\hat{\rho}_{m,u}^{(k)}(t_j)$ is an estimate of $r_{m,u} \cdot \underline{1}^{(k)} + b_{m,u}$ at time t_j . This recursion can be initialized using the first differential code measurement ($\Delta\hat{\rho}_{m,u}^{(k)}(t_0) = \Delta\tau_{m,u}^{(k)}(t_0)$).

The first term of the recursion given by (5) is the current differential code measurement weighted by $1/L$. If the recursion is updated every second, then L is generally set between 100 and 200. The second term of the recursion is the current best estimate of pseudorange propagated forward by the carrier phase difference. If the carrier phase tracking loops have held lock, then the carrier phase difference estimates the change in the pseudorange with very high accuracy. The forward prediction does not suffer from any integer ambiguity because only the change in the carrier measurement is used. If the carrier phase tracking loops do suffer from a cycle slip, then the filter must be reset. The averaging time is usually limited to 100–200 s, so that re-initialization of the filter does not cause a lengthy outage.

After smoothing, the estimated pseudoranges replace the raw code phase measurements in (3). The accuracy of the estimated position is generally improved because the smoothing has attenuated any error sources which have decorrelation times less than 100 s.

D. Carrier Phase Positioning

A time series of differential carrier observations can also be processed to estimate position directly [21], [26]. With seven satellites, the carrier measurements will provide accuracy better than one meter within 200 s or so. In principle, code phase measurements are not required for this convergence. However, a position estimate based on differential code phase measurements does provide an excellent initial position for the iteration.

At time t_1 , the carrier differences are described by the following system of equations:

$$\Delta\phi(t_1) = \begin{bmatrix} \Delta\phi_{m,u}^{(k1)}(t_1) \\ \Delta\phi_{m,u}^{(k2)}(t_1) \\ \vdots \\ \Delta\phi_{m,u}^{(K)}(t_1) \end{bmatrix} = \begin{bmatrix} \underline{1}^{(k1)}(t_1) & 1 & 0 \cdots 0 \\ \underline{1}^{(k2)}(t_1) & 1 & \\ \vdots & \vdots & \\ \underline{1}^{(K)}(t_1) & 1 & I \end{bmatrix} \cdot \begin{bmatrix} r_{m,u}(t_1) \\ b_{m,u}(t_1) + \Delta N_{m,u}^{(k1)}\lambda \\ \Delta N_{m,u}^{(k1,k2)}\lambda \\ \vdots \\ \Delta N_{m,u}^{(k1,K)}\lambda \end{bmatrix} + \Delta\nu_\phi(t_1) = \tilde{G}(t_1)\tilde{x}(t_1) + \Delta\nu_\phi(t_1).$$

As shown, the observations can be written as a matrix which is determined by satellite geometry alone ($\tilde{G}(t_1)$) times a state vector ($\tilde{x}(t_1)$) plus observation noise ($\Delta\nu_\phi(t_1)$). The state vector contains the estimanda $r_{m,u}(t_1) = [x_{m,u}(t_1) \ y_{m,u}(t_1) \ z_{m,u}(t_1)]^T$ as well as nuisance parameters due to the clock offset and cycle ambiguities.

The cycle ambiguities can be eliminated by subtracting the carrier phase measurements at two adequately separated times as follows:

$$\begin{aligned} \Delta\phi(t_2) - \Delta\phi(t_1) &= \tilde{G}(t_2)\tilde{x}(t_2) - \tilde{G}(t_1)\tilde{x}(t_1) \\ &\quad + \Delta\nu_\phi(t_2) - \Delta\nu_\phi(t_1) \\ &= G(t_2)\underline{x}(t_2) - G(t_1)\underline{x}(t_1) \\ &\quad + \Delta\nu_\phi(t_2) - \Delta\nu_\phi(t_1) \\ &= G(t_2)(\underline{x}(t_1) + \delta\underline{x}) - G(t_1)\underline{x}(t_1) \\ &\quad + \Delta\nu_\phi(t_2) - \Delta\nu_\phi(t_1) \\ &= \delta G\underline{x}(t_1) + G(t_2)\delta\underline{x} \\ &\quad + \Delta\nu_\phi(t_2) - \Delta\nu_\phi(t_1) \end{aligned} \quad (6)$$

where G and $\underline{x}$ denote the reduced observation matrix and state after the subtraction has removed the nuisance integers. In other words

$$\begin{aligned} G(t)\underline{x}(t) &= \begin{bmatrix} \underline{1}^{(k1)}(t) & 1 \\ \underline{1}^{(k2)}(t) & 1 \\ \vdots & \vdots \\ \underline{1}^{(K)}(t) & 1 \end{bmatrix} \begin{bmatrix} x_{m,u}(t) \\ y_{m,u}(t) \\ z_{m,u}(t) \\ b_{m,u}(t) \end{bmatrix} \\ \delta G &= G(t_2) - G(t_1) \\ \delta\underline{x} &= \underline{x}(t_2) - \underline{x}(t_1). \end{aligned} \quad (7)$$

The last equation in (6) is key. It shows that the measurements are sensitive to the change in the location of the aircraft ($\delta\underline{x}$) through the matrix $G(t_2)$. As long as the satellite geometry is favorable at time t_2 , then the change in aircraft position can be determined with high accuracy. In essence, the carrier measurements provide an estimated trajectory for the aircraft, but that trajectory has an unknown starting point.

The starting point for the aircraft trajectory only reveals itself as the satellites move. In other words, the measurements are sensitive to the starting location $\underline{x}(t_1)$ only

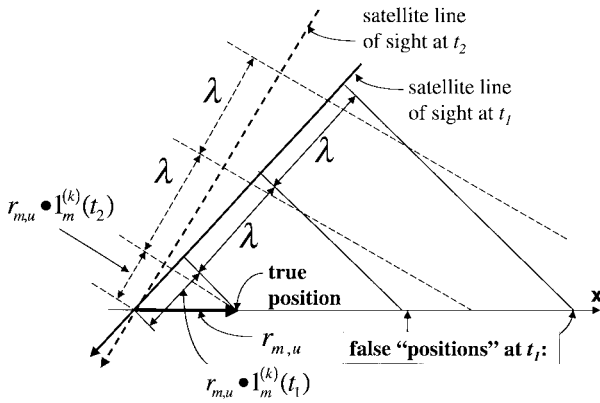


Fig. 9. Geometric diversity. The motion of the satellite renders the wavelength ambiguity observable in this one dimensional position fixing problem.

through $\delta G = G(t_2) - G(t_1)$. Consequently, a significant change in geometry is required to render $\underline{x}(t_1)$ observable. If the line of sight vectors sweep a significant angle, then geometric diversity is achieved and accurate position estimates can be generated.

Geometric diversity is depicted in Fig. 9, which shows a one-dimensional position-fixing problem—assume $y_{m,u}$, $z_{m,u}$, and $b_{m,u}$ are known and that the measurements are noiseless. In this case, the measurements at times t_1 and t_2 are

$$\begin{aligned}\Delta\phi_{m,u}^{(k)}(t_1) &= \underline{x}_{m,u} \cdot \hat{1}_m^{(k)}(t_1) + \Delta N_{m,u}^{(k)} \lambda \\ \Delta\phi_{m,u}^{(k)}(t_2) &= \underline{x}_{m,u} \cdot \hat{1}_m^{(k)}(t_2) + \Delta N_{m,u}^{(k)} \lambda\end{aligned}\quad (8)$$

since we have also assumed that the user is stationary and does not suffer any cycle slips. Each carrier phase measurement defines a number of possible positions along the x axis because of the wavelength ambiguities. However, the false positions reveal themselves provided the difference $\hat{1}_m^{(k)}(t_1) - \hat{1}_m^{(k)}(t_2)$ is large enough to render the estimanda $\underline{x}_{m,u}$ observable. Only the true solution remains stationary as the line of sight vector follows the satellite across the sky.

In general, least squares techniques can be used to iteratively solve (6) for the change in user position and clock ($\delta \underline{x}$) as well as the initial value of the state $\underline{x}(t_1)$. Specifically

$$\begin{aligned}\delta \hat{\underline{x}} &= \text{Pseudoinverse}\{G(t_2)\} \\ &\quad \cdot [\Delta\phi(t_2) - \Delta\phi(t_1) - \delta G \underline{x}(t_1)] \\ \hat{\underline{x}}(t_1) &= \text{Pseudoinverse}\{\delta G\} \\ &\quad \cdot [\Delta\phi(t_2) - \Delta\phi(t_1) - G(t_2)\delta \underline{x}].\end{aligned}\quad (9)$$

A similar iteration is explored by Van Graas and Lee in [26]; they show that such an iteration converges reliably. The convergence time depends on the number of satellites in view and the satellite geometry.

While estimating the position at $t_1(\hat{\underline{x}}(t_1))$, the measurement must be adjusted by $G(t_2)\delta \underline{x}$. This adjustment uses the best current estimate of $\delta \underline{x}$. While estimating the change in position ($\delta \hat{\underline{x}}$), the measurements must be adjusted for the change in measurement geometry ($\delta G \underline{x}(t_1)$). This adjustment uses the current best estimate of $\underline{x}(t_1)$, where

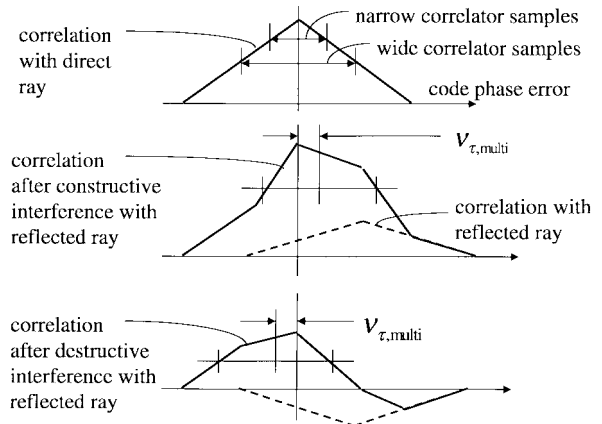


Fig. 10. Correlation peak without multipath, with constructive interference from a reflected ray and with destructive interference from a reflected ray.

this estimate may come from any source of position information. Of course, the differential code measurements are the most logical source for an estimate of $\underline{x}(t_1)$. In this case, the differential code and carrier measurements are being combined in the position domain. This contrasts the approach described in Section II-C, where they were combined in the measurement domain.

III. PERFORMANCE OF CARRIER-AIDED CODE POSITIONING

Carrier aiding is effective against noise and interference, because these perturbations are white—their decorrelation times are small compared to the averaging time enabled by carrier aiding. However, carrier smoothing is not always effective against GPS multipath, which can be bias-like for 100 s or more. This situation is discussed in Section III-A.

A. Multipath Effect on Carrier-Smoothed Code Phase Measurements

As depicted in Fig. 10, code multipath introduces an error into code phase measurements by distorting the peak of the correlation function. Most receivers use a delay lock loop (DLL) to estimate the phase of the received code [27], [28]. This loop attempts to straddle the peak of the correlation function with an early sample on the leading edge and a late sample on the falling edge. The loop strives to track the peak of the correlation function by slewing the samples such that their difference is nulled.

As shown in the top trace of Fig. 10, the correlation peak is symmetric in the absence of multipath, but the arrival of multipath upsets this vital symmetry. Depending on the relative phase of the two rays, the correlation function for the reflected ray may be added to or subtracted from the correlation function for the direct ray. If the reflected ray is in-phase with the direct ray, then constructive interference results in composite correlation functions like the one shown in the middle trace of Fig. 10 where multipath delays the estimated code phase. If the interference is destructive, then multipath results in an advance in the estimated code phase.

DLL's are characterized by their correlator spacing—the time between the early and late samples. Until recently, most GPS receivers have used the wider correlator spacing shown in Fig. 10 with $d = 1.0$. In other words, the correlator spacing is approximately equal to one chip width or $1 \mu\text{s}$. As will be discussed shortly, the variance of the error due to noise is proportional to the correlator spacing [28]. Consequently, recent GPS receivers have started to use correlator spacing approximately equal to five or ten percent of a chip width ($d = 0.05$ or $d = 0.1$). Little gain is realized from making the spacing any smaller, because the correlation peak is rounded by the finite bandwidth of the satellite and receiver. For some situations, reduced correlator spacing also mitigates multipath.

GPS multipath is characterized by its Doppler frequency and delay relative to the direct ray. These characteristics are shown in Figs. 7 and 8. As shown in Fig. 7, multipath Doppler is the time rate of change of the relative delay ($\Delta\dot{r}_{MD}$) when the delay is measured in radians. In other words

$$\dot{\theta}_{MD} = \frac{2\pi}{\lambda} \Delta\dot{r}_{MD} \quad \text{rad/s} \quad (10)$$

where λ is the GPS wavelength of 19 cm. If the Doppler frequency is high relative to the loop bandwidth, then the multipath will be attenuated by the carrier smoothing. For example, aircraft multipath is attenuated significantly by carrier smoothing, because the airframe motion rapidly changes the relative path length and whitens the multipath. However, ground reflections can generate multipath with very low Doppler frequencies. These errors persist as biases and are transmitted to the aircraft as errors in the ground measurements.

The ground multipath depicted in Fig. 7 is particularly worrisome. As shown, the GPS signal consists of a direct ray and a ray which is reflected from beneath the ground. This reflected ray is attenuated by the reflection and by the directivity of the reference antenna. However, it can still be troublesome because its Doppler frequency is very low. Specifically, its delay and Doppler relative to the direct ray are approximately given by

$$\begin{aligned} \Delta r_{MD} &\approx 2h_m \sin \epsilon l^{(k)} \quad \text{m} \\ \theta_{MD} &\approx \frac{2\pi}{\lambda} 2h_m \sin \epsilon l^{(k)} \quad \text{rad} \\ \dot{\theta}_{MD} &\approx \frac{2\pi}{\lambda} 2h_m \cos \epsilon l^{(k)} \frac{d\epsilon l}{dt} \quad \text{rad/s} \end{aligned} \quad (11)$$

where h_m is the reflection depth in meters. The elevation angle of the satellite and its time rate of change are given by $\epsilon l^{(k)}$ and $d\epsilon l/dt$, respectively. Approximations (11) are certainly valid for reflection depths of 100 m or less.

Fig. 11 shows the delay of the reflected ray relative to the direct ray for the multipath scenario of Fig. 7. It shows delay as a function of elevation angle with reflection depth as a parameter. Fig. 12 is an upper bound on the relative Doppler for the same scenario. (See Appendix B for the derivation of Fig. 12.) It shows a Doppler bound as a function of elevation angle and also uses reflection depth as

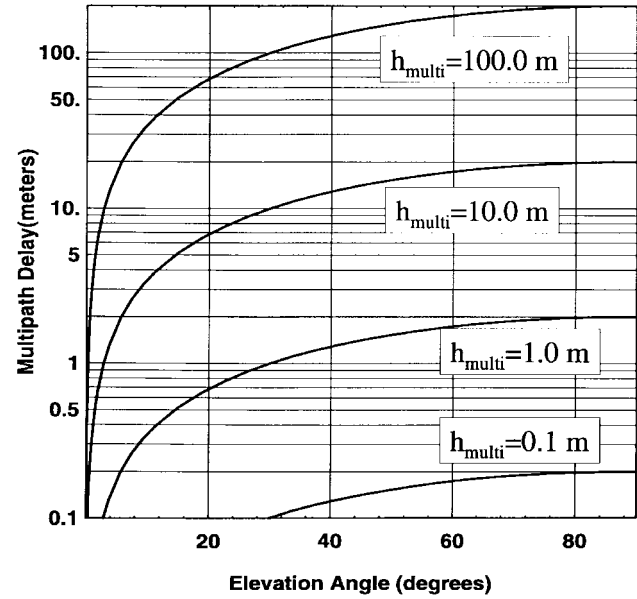


Fig. 11. Multipath delay versus elevation angle of the GPS satellite. Multipath is assumed to be reflected from below the reference antenna and the reflection depth is a parameter in the figure.

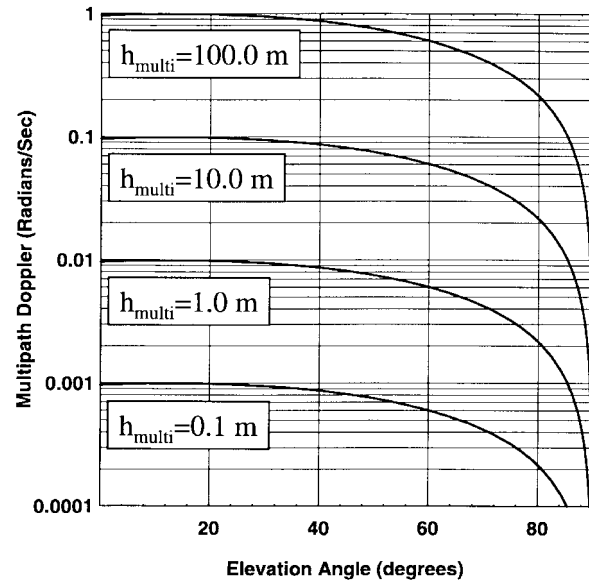


Fig. 12. Upper bound for multipath Doppler versus elevation angle of the GPS satellite. Multipath is assumed to be reflected from below the reference antenna and the reflection depth is a parameter in the figure. The loop bandwidth for 100 s of carrier smoothing is approximately 0.03 rad/s.

a parameter. The curves give the exact Doppler only for the case where the satellite rises perpendicular to the horizon. For smaller reflection depths, even the upper bound is small compared to the one-sided loop bandwidth of 0.03 rad/s enabled by carrier aiding. Consequently, reference station multipath must be treated as a bias.

The size of the code phase bias due to multipath is analyzed in Appendix C, which yields the bounds shown in Fig. 13. As the relative phase of the multipath varies, the code phase error swings between the upper and lower

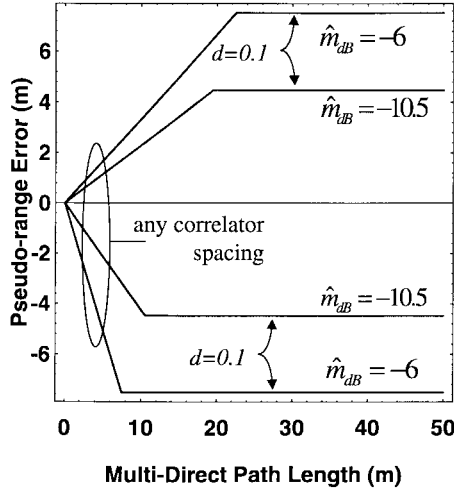


Fig. 13. Bounds on code phase error due to multipath. The actual error varies between the indicated bounds as the relative phase changes.

bounds shown in the figure. For larger delays, the errors are proportional to the multipath amplitude and the correlator spacing, but they are independent of delay. For small delays, the error bounds are proportional to delay and grow with the multipath amplitude, but they are independent of the correlator spacing. In fact, the multipath error bound for small delays is

$$|\nu_{\tau, \text{multi}}| \leq \Delta r_{M,D}(e\ell^{(k)}) \frac{\hat{m}}{1 - \hat{m}} \text{ m.} \quad (12)$$

The tolerable multipath amplitude as a function of the multipath delay and tolerable pseudorange error is

$$\hat{m}_{\text{dB}} \leq 20 \log_{10} \left[\frac{|\nu_{\tau, \text{multi}}|}{|\nu_{\tau, \text{multi}}| + \Delta r_{M,D}(e\ell^{(k)})} \right]. \quad (13)$$

This relationship is plotted in Fig. 14, which assumes that the pseudorange error due to multipath must be limited to 0.1 m ($|\nu_{\tau, \text{multi}}| \leq 0.1$ m). The small GPS antennas, used in mobile applications, cannot provide the attenuation required by Fig. 14. However, the antennas described in [29] or [30] can provide such a gain roll off with elevation angle.

B. Noise Effect on Carrier-Smoothed Code Phase Measurements

Clearly, carrier aiding will be effective against noise and interference, because these error mechanisms decorrelate more quickly than 100 s. In the presence of noise and interference, the early-power minus late-power DLL has an error variance given by [28]

$$\sigma_{\tau}^2(C_X) = \frac{d(cT_C)^2 BW_{L,1\text{-sided}}}{2 \frac{C_X(e\ell^{(k)})}{N_0 + I_0}} \cdot \left[1 + \frac{2}{(2-d) \frac{C_X(e\ell^{(k)})}{N_0 + I_0} T_{50}} \right] \text{ m}^2 \quad (14)$$

where $T_C = 1 \mu\text{s}$ is the chip duration for the C/A code and c is the speed of light. In addition, $C_X(e\ell^{(k)})/(N_0 + I_0)$

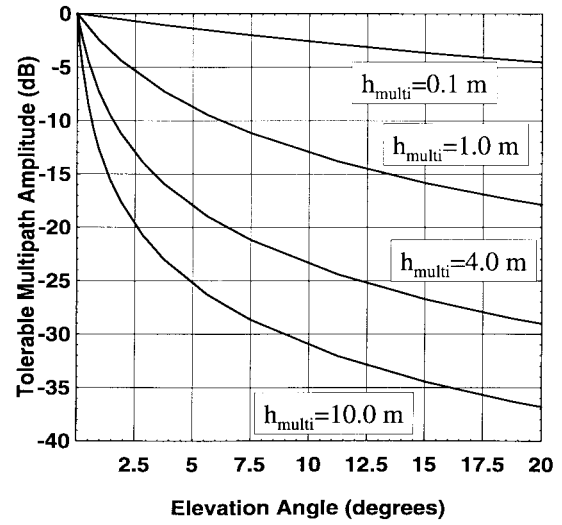


Fig. 14. Tolerable multipath amplitude (dB) versus elevation angle of the GPS satellite. Multipath is assumed to be reflected from below the reference antenna and the reflection depth is a parameter in the figure. The tolerable pseudorange error due to multipath is assumed to be 0.1 m, and the multipath amplitude is normalized to the amplitude of the direct ray.

is the received signal power divided by the noise power spectral density (N_0) plus the equivalent interference power spectral density (I_0). For GPS, the received signal power is a function of the satellite's elevation angle ($e\ell^{(k)}$). The one-sided bandwidth of the loop filter is denoted $BW_{L,1\text{-sided}}$. If carrier aiding is used with a time constant of $T = 100$ s, then this bandwidth is $1/2T = 0.005$ Hz ≈ 0.03 rad/s.

In (14), d is the correlator spacing of the DLL. As shown, reducing the correlator spacing from $d = 1.0$ to $d = 0.1$ decreases the noise variance by a factor of 10. Hence, narrow correlators do provide a significant advantage against noise—even though they are only partially effective against LAAS multipath.

Since the GPS signals are modulated by data, DLL's must employ some form of nonlinear processing to strip the data. This nonlinear operation inevitably gives rise to squaring loss. The squaring loss for the early-power minus late-power DLL is inside the square brackets of (14), where $T_{50} = 1/50$ s is the duration of the GPS data bit.

C. Covariance Analysis for Carrier-Smoothed Code

A covariance analysis can be used to approximate the vertical accuracy provided by carrier-smoothed code. This analysis assumes that the vertical position error is given by (4). The covariance matrix ($P_{\Delta\tau}$) is assumed to be diagonal, because measurement errors for different satellites are nearly uncorrelated. The diagonal elements of the covariance matrix are given by

$$P_{\Delta\tau, k, k} \approx 2 \left[\sigma_{\tau}^2 \left(C(e\ell^{(k)}) \right) + \frac{(0.1)^2}{3} \right] + \sigma_{\text{tropo,sv}}^2(e\ell^{(k)}, \Delta h_u) \quad (15)$$

$$\sigma_{\text{tropo,sv}}(e\ell^{(k)}, \Delta h_u) \approx \frac{\sigma_N \Delta h_u \times 10^{-6}}{\sin(e\ell^{(k)})}$$

where $P_{\Delta\tau,k,k}$ is the element in the k th row and k th column of the covariance matrix.

The variance due to noise and multipath appears inside the square brackets. These errors are assumed to be equal and uncorrelated from ground to air. The noise variance is from (14) where the received signal power is assumed to be a smooth function of elevation angle with $C_X(90^\circ)/(N_0 + I_0) = 45$ dB-Hz, and $C_X(5^\circ)/(N_0 + I_0) = 35$ dB-Hz.

The variance of the multipath error is set equal to $(0.1)^2/3$ for both the air and ground. In general, the multipath error will also be a function of elevation angle, but the dependence is not obvious. At low-elevation angles, the reflection is harder to attenuate with the antenna pattern, but the delays are shorter.

The tropospheric error model comes from [28] and assumes that the error is due to the altitude of the aircraft relative to the reference receiver (Δh_u) and uncertainty in the surface refractivity (σ_N). The analysis described below assumes that the aircraft is only 30 m above the reference station, and that the error in the estimated refractivity is 100. The tropospheric modeling error will be correlated from satellite to satellite. However, we ignore this correlation because the tropospheric error at an altitude of 30 m is relatively small.

Figs. 2 and 3 plot the resulting vertical accuracy as a function of time for an aircraft approaching San Francisco International Airport. As shown, differential GPS with carrier-aided code measurements provides much better accuracy than the SPS without differential corrections. However, this improvement is achieved by reducing the measurement noise and not by increasing the number of measurements. Hence, the resulting accuracy is still sensitive to the geometry of the GPS constellation. In particular, Fig. 3 shows a spike in the middle of the day where too few signals are available. The pseudolites discussed in Section IV address this shortcoming, but first we review some flight data for a carrier-aided code-phase landing system.

D. Experimental Results and Current Work

Landing systems based on carrier-smoothed code have been prototyped by Wilcox Electric [31] and Ohio University [26]. The Wilcox system uses narrow correlator technology to mitigate multipath and noise. In addition, their prototype has the same aircraft interface as existing instrument landing systems (ILS), and so the airborne receiver would be a drop-in replacement for today's ILS receiver. This strategy is aimed at reducing the need to re-certify other elements in the aircraft.

The Wilcox prototype was evaluated at the FAA Technical Center in April of 1995. These trials used a Boeing 727-200 provided by FEDEX. This aircraft does not utilize an inertial system and the LAAS solution was not enhanced in any way.

Fig. 15(a) shows the difference between the vertical position of the aircraft as reported by the Wilcox system and the FAATC laser tracker. In other words, the figure shows the vertical navigation sensor error (NSE) for the GPS-

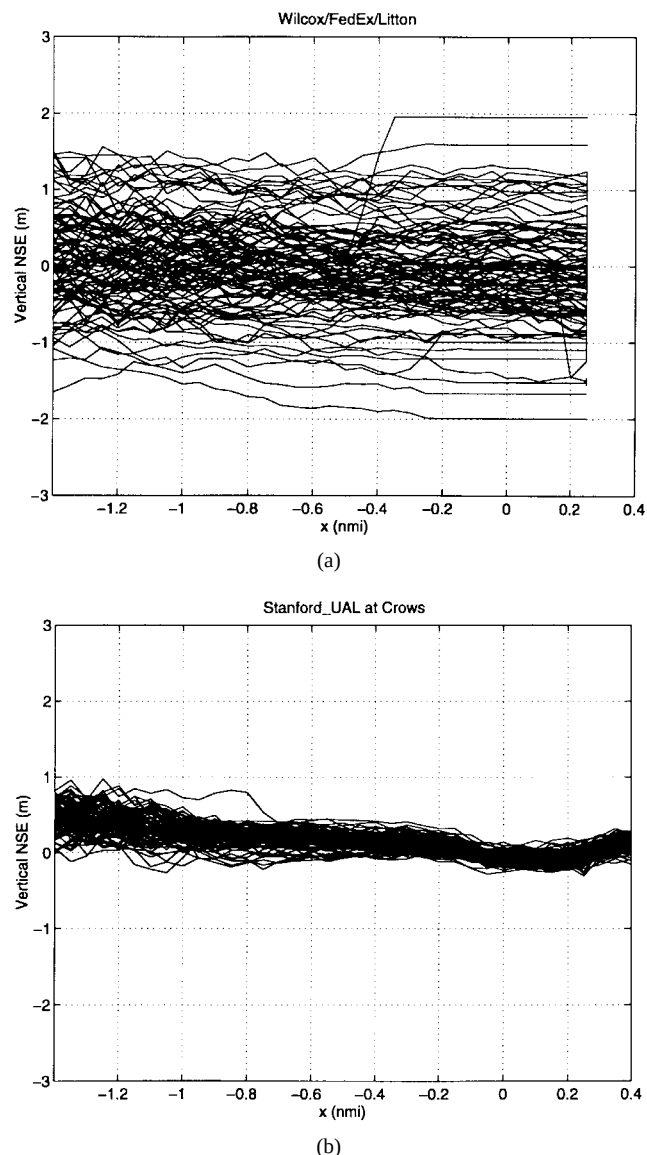


Fig. 15. Vertical navigation sensor error from flight trials conducted by Wilcox Electric at the FAA Technical Center and Stanford University at Crow's Landing. Wilcox used carrier-aided code-phase and narrow correlation technology. Stanford University used a differential carrier phase system and their results are presented to give an approximate notion of the accuracy of a laser tracker.

based landing system as a function of the aircraft location along the approach path. (The desired glide path for the aircraft intersects the runway at $x = 0$ nm.)

Fig. 15(b) approximately establishes the accuracy of the laser tracker. It shows the difference between a similar laser tracker and an extremely accurate truth system based on the GPS carrier phase [33]. The differences in Fig. 15(a) can be attributed to errors in the carrier-aided code-phase solution because they are significantly larger than the differences shown in Fig. 15(b).

The vertical NSE measured at the FAATC is greater than that predicted by our covariance analysis and shown in Fig. 2. Moreover, the errors vary slowly—they are bias like for the duration of the approach. As such, they are

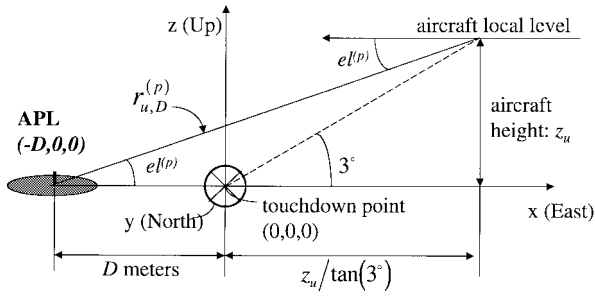


Fig. 16. APL's showing the basic measurement geometry used for analyzing APL's.

probably due to multipath near the reference antenna. Even though Wilcox used reasonable care in antenna selection and placement, their reference antenna was not optimized for multipath mitigation.

IV. APL's

As depicted by Figs. 5 and 16, APL's are ground transmitters placed on the airport property to augment the availability of LAAS. Three distinct kinds of pseudolites have been considered as part of the LAAS development: ranging APL's, hyperbolic APL's, and integrity beacons. Of these, ranging APL's are the simplest and have been described most extensively in the literature [34]. Hyperbolic APL's are a very recent invention with the first descriptions in [35] and [36]. Finally, integrity beacons are well described in [16] and [33]. They provide remarkable accuracy and integrity, but the integrity beacons are pseudolites installed under the glide path. As such, they require the maintenance and operation of equipment located off airport property. For this reason, integrity beacons are not being pursued currently, and this section only addresses ranging APL's and hyperbolic APL's. Prior to these discussions, APL signal design is described. At the end of this section, we discuss some of the siting issues associated with APL's.

A. APL Signal Design

Pseudolite signal designs have received a fair amount of attention in the literature [34], [37], [38] where the designs are directed at solving the near/far problem associated with pseudolites. After all, the power received from the GPS satellites is nearly independent of aircraft location—received signal power typically decreases by only 10 dB as the satellite moves from zenith to within 5° of the horizon. In contrast, the power received from a pseudolite will change by 60 dB from an acquisition range of 10 km to a point of closest approach on the runway of 10 m. The qualitative discussion found here will only introduce the subject, since the best signal design for LAAS is still under discussion.

The APL signal is pulsed to reduce interference to GPS receivers. The pulsing duty cycle may be 10% ($\eta = 0.10$) with pulse durations of 100 μ s or so. Such a short pulse will have little effect on most GPS receivers, since they hard limit, clip, or blank the incoming signal. For this

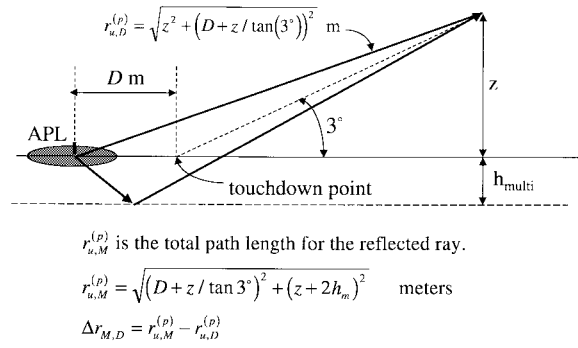


Fig. 17. APL multipath. Each APL generates a direct ray to the aircraft and a reflected ray, where the reflection occurs h_m meters below the APL antenna.

reason, the full APL signal energy is not integrated into the receiver's various decision statistics. In most cases, the APL pulse simply prevents the accumulation of the GPS signal energy when it is on. However, some GPS receivers employ very fast automatic gain control (AGC). If the AGC is fast compared to the duration of the pulse, then larger degradations can result.

Pulsing alone may not guarantee noninterference. In this case, the APL could use a 10 MHz chipping rate rather than the 1 MHz chipping rate used by the SPS [38]. This option is attractive because the GPS P code already uses a code rate of 10 Mhz and so appropriate receiver hardware exists.

Alternatively, the APL signal could be offset in frequency relative to the GPS L1 frequency. Offsets of 8 MHz or more would be reassuring, because the filters in the receiver front end would partially attenuate the APL signal. However, the frequency choice must also reflect the following considerations. First, frequencies near or within the GPS band may be converted by any given receiver's local oscillator to fall at L1. If such image interference is troublesome, the sought-after protection of the GPS signals will not be realized. Second, frequencies outside of the GPS band will require coordination with the users of those bands. For example, GLONASS—the Russian satellite navigation system—uses the frequencies from 1590–1610 MHz. Third, larger offsets increase the interchannel bias introduced by the front end of the receiver. This bias is the difference in the phase and group delays of the receiver at the GPS and APL frequencies.

In summary, the near/far problem is probably best solved by pulsing and using a higher chipping rate.

B. Ranging APL's

Ranging APL's simply augment the aircraft's navigation solution with another ranging signal from the ground. Fundamentally, the new signal is processed like the satellite signals—the carrier is used to smooth the code.

However, multipath generated by the APL broadcast is now a more difficult consideration. As shown in Figs. 16 and 17, the aircraft approaches along a glide path which only makes an angle of 3° with the airport surface. If the APL is located next to the desired touchdown point ($D = 0$), then the elevation angle ($e l^{(p)}$) is $-3°$. If the APL is

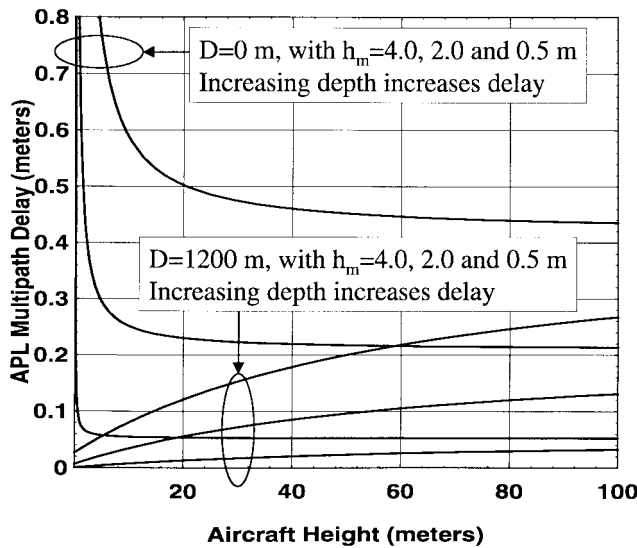


Fig. 18. APL multipath delay versus aircraft height. The figure shows the arrival time delay for a ray reflected below the APL transmit antenna. The delay is measured relative to the arrival time of a direct ray.

down the runway from the touchdown point, then elevation angles of -1° or -2° could well be commonplace. At these low angles, a strong reflection is likely.

Fortunately, the reflected ray will have a very small delay relative to the direct. The multipath delay is shown in Fig. 18 as a function of aircraft height. Fig. 18 includes reflection depth (h_m) as a parameter, and considers an APL located next to the touchdown point ($D = 0$) as well as an APL 1200 m down the runway ($D = 1200$). When the aircraft is at an altitude of 20 m or more, the multipath delay is always less than 0.5 m. Below 20 m, the aircraft will depend on its radar altimeter. According to (12), the resulting pseudorange error will always be smaller than 0.5 m, provided the multipath amplitude is bounded by 0.5. Such multipath suppression is certainly possible using the antennas described in [29] and [30]. In the analysis that follows, the multipath error (in meters) is assumed to follow a uniform distribution over the interval $[-0.5, 0.5]$.

A covariance analysis can be used to illustrate the efficacy of augmentation using ranging APL's. This analysis assumes that the vertical position error is given by (4). However, it augments the traditional GPS geometry matrix (G) with the following row for the added signal:

$$G^{(p)} = \begin{bmatrix} \frac{\partial \Delta\tau}{\partial x} & \frac{\partial \Delta\tau}{\partial y} & \frac{\partial \Delta\tau}{\partial z} & \frac{\partial \Delta\tau}{\partial b} \end{bmatrix} \\ = \begin{bmatrix} \sin az^{(p)} \cos el^{(p)} & \cos az^{(p)} \cos el^{(p)} & \sin el^{(p)} & 1 \end{bmatrix}.$$

The azimuth and elevation angles made by the line of sight vectors from the aircraft to the pseudolite are denoted $az^{(p)}$ and $el^{(p)}$, respectively. (Azimuth is traditionally measured from north and elevation is positive up from the horizontal.)

Once again, the satellite measurement errors are assumed to be independent from satellite to satellite and include

noise, multipath, and tropospheric modeling errors. The variance of the error in a corrected satellite measurement is given by

$$P_{\Delta\tau,k,k} = 2 \left[\frac{\sigma_\tau^2 [C(el^{(k)})]}{(1-\eta)} + \frac{(0.1)^2}{3} \right] + \sigma_{\text{tropo,sv}}^2 (el^{(k)}, \Delta h_u). \quad (16)$$

The variance of the error in the corrected APL measurement is given by

$$P_{\Delta\tau,K+1,K+1} = \frac{2\sigma_\tau^2 [C^{(\text{APL})}]}{\eta} + \frac{(0.5)^2}{3} + \sigma_{\text{tropo,APL}}^2 (r_u^{(p)}) \\ \sigma_{\text{tropo,APL}} (r_u^{(p)}) \approx \sigma_N r_u^{(p)} \times 10^{-6} \\ = \frac{\sigma_N \Delta h_u \times 10^{-6}}{-\sin(el^{(p)})} \text{ m.} \quad (17)$$

As before, the noise variances ($\sigma_\tau^2(C)$) are given by (14), where $C(el^{(k)})$ is the peak power of the k th satellite. The peak power of the APL signal ($C^{(\text{APL})}$) is assumed to be equal to the satellite power when the aircraft is 10 km from the airport and grows as the aircraft approaches in accord with the $1/r^2$ propagation law.

The effect of pseudolite pulsing is shown in both (16) and (17), where the pseudolite duty cycle is $0 < \eta < 1$. The satellite noise variance increases with η , because integration of the satellite signal power is suspended by receiver blanking when the pseudolite is on. The APL noise variance decreases with η , because energy from the APL signal can only be collected when the pseudolite is transmitting.

As before, the error due to satellite multipath is assumed to have a variance of $(0.1)^2/3$. APL multipath is assumed to have a uniform distribution over $[-0.5, 0.5]$ and so it has a variance of $(0.5)^2/3$.

The tropospheric error model for the satellites is the same as we used earlier in (15). The tropospheric error model for the APL also comes from [28]. It is a function of the error in the estimated surface refractivity (σ_N). The tropospheric modeling error also grows with the distance from the pseudolite to the aircraft ($r_u^{(p)} = -\Delta h_u / \sin el^{(p)}$). The analysis described below assumes that the aircraft is only 30 m above the pseudolite, and that the error in the estimated refractivity is 100.

The results of our covariance analysis for an aircraft approaching San Francisco are shown in Figs. 2 and 3. As shown in Fig. 3, the ranging APL does much to mitigate the outages suffered by the GPS constellation operating without augmentation.

C. Hyperbolic APL's

Two or more APL's can be configured to provide a fixed set of hyperbolic lines of position. The carrier phase of at least two transmissions are synchronized, and the aircraft measures the difference in the received carrier phases. This

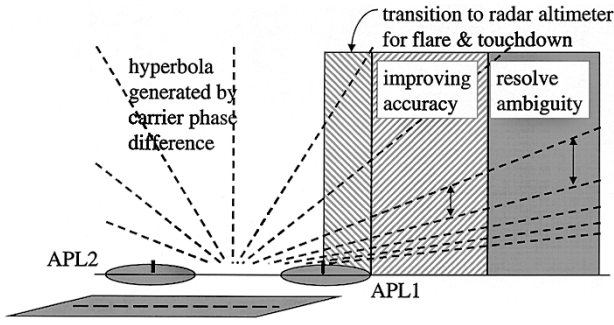


Fig. 19. In-track APL's shown along with the hyperbola defined by constant differences in the received APL carrier phases. The hyperbola are the dashed curves.

difference is given by

$$\begin{aligned}\Delta\phi^{(1,2)} &= r_u^{(p1)} - r_u^{(p2)} + \Delta N\lambda + \Delta\nu \\ &= \Delta r + \Delta N\lambda + \Delta\nu\end{aligned}\quad (18)$$

where $r_u^{(p)}$ is the range from the user to either of the APL's. This carrier phase difference defines a set of hyperbola, where each hyperbola corresponds to one value of the integer ambiguity ($\Delta N\lambda$). These lines of position are exactly analogous to the hyperbola formed by Loran-C time differences [39], and they have some fortunate properties.

First, the lines of position are relatively far apart when the aircraft is far from the airport. Consequently, the carrier-smoothed code position fix can be used to resolve the integer ambiguity ($\Delta N\lambda$) before the critical portion of the approach has begun. Second, the lines of position converge as the aircraft moves down the glide slope—providing better vertical accuracy as position quality becomes more critical.

In this section, we will analyze the hyperbolic APL's shown in Fig. 19. These APL's are placed in track alongside any runway for which a Category II or III precision approach capability is desired [35]. One is placed close to the touchdown point and the second is placed some 1000–2000 m farther down the runway. This in-track configuration is only one of many interesting possibilities.

1) *Sensitivity Analysis:* The differential carrier phase measurement has the following sensitivities to perturbations in the user's estimated position:

$$\begin{aligned}\frac{\partial\Delta\phi}{\partial x} &= \frac{x+D}{r^{(p2)}} - \frac{x+0}{r^{(p1)}} \\ &= \sin az^{(p2)} \cos el^{(p2)} - \sin az^{(p1)} \cos el^{(p1)} \\ \frac{\partial\Delta\phi}{\partial y} &= \frac{y}{r^{(p2)}} - \frac{y}{r^{(p1)}} \\ &= \cos az^{(p2)} \cos el^{(p2)} - \cos az^{(p1)} \cos el^{(p1)} \\ \frac{\partial\Delta\phi}{\partial z} &= \frac{z}{r^{(p2)}} - \frac{z}{r^{(p1)}} = \sin el^{(p2)} - \sin el^{(p1)}\end{aligned}\quad (19)$$

where we continue to use the coordinate system introduced in Fig. 16. Both APL's are placed on the x axis, with one next to the touchdown point (0, 0, 0) and the second placed D m west of the touchdown point ($-D$, 0, 0). Elevation is positive above the x - y plane and azimuth is measured clockwise from north.

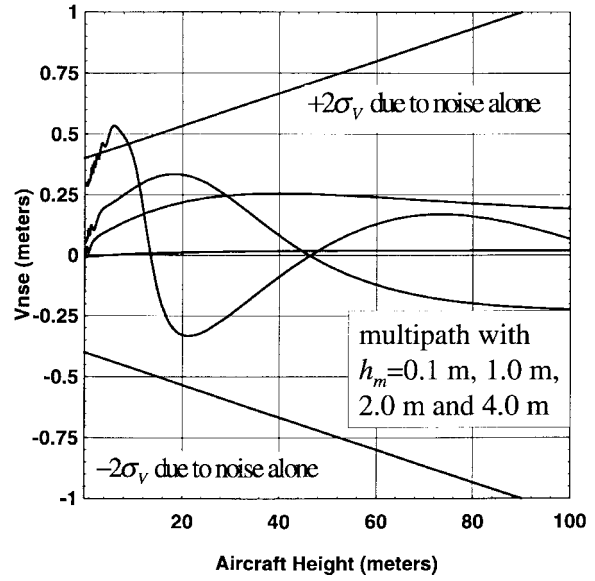


Fig. 20. Vertical accuracy along the glide slope provided by in-track APL's alone. The figure shows the vertical error due to noise-alone and multipath-alone for a variety of reflection depths. The multipath results assume that both APL antennas have gains which roll off by 2.5 dB per degree of elevation.

If the aircraft is approaching from the east on a 3° glide slope, then

$$\begin{aligned}G^{(p1)} &= \begin{bmatrix} \frac{\partial\Delta\phi}{\partial x} & \frac{\partial\Delta\phi}{\partial y} & \frac{\partial\Delta\phi}{\partial z} & \frac{\partial\Delta\phi}{\partial b} \end{bmatrix} \\ &\approx \begin{bmatrix} 0 & 0 & \left(\frac{z}{D+20z} - \frac{1}{20} \right) & 0 \end{bmatrix} \\ &\xrightarrow{D \rightarrow \infty} \begin{bmatrix} 0 & 0 & -\frac{1}{20} & 0 \end{bmatrix}.\end{aligned}\quad (20)$$

As shown, the differential phase measurement is insensitive to changes in the horizontal position of the aircraft. It also appears to have low sensitivity to the critical vertical position. However, the differential carrier phase measurement is very precise with standard deviations of approximately 1 cm due to noise. The corresponding vertical position error ($\pm 2\sigma_v$) is plotted in Fig. 20 for $D = 1200$ m and 1 cm of measurement error. As shown, the vertical navigation sensor error at an aircraft height of 30 m will be approximately 0.6 m. This vertical information from the in-track APL's will be a welcome addition to the carrier-smoothed code solution.

At greater distances, the sensitivity is such that the error due to an incorrect cycle is large compared to the vertical accuracy of a carrier-smoothed code solution. For example if the aircraft is 10 km from the airport at an approximate altitude of 500 m and the APL's are spaced by $D = 1200$ m, then a measurement error of one wavelength (19 cm) yields a vertical position error of 35 m. Clearly, the accuracy of a carrier-smoothed code solution will be able to resolve this wavelength ambiguity.

2) *Carrier Phase Multipath:* As with ranging APL's, multipath is an important concern. Fortunately, the differential carrier phase measurement is much less sensitive to multipath than the code phase measurement.

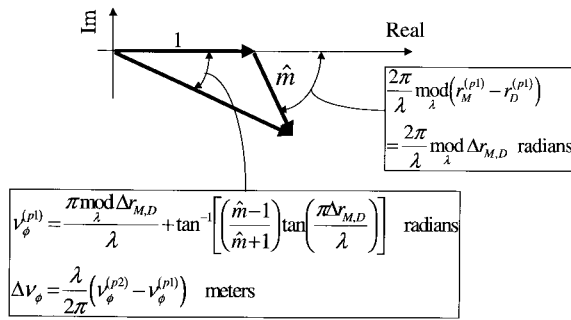


Fig. 21. Carrier phase multipath for one in-track APL. The carrier phase measurement is much less sensitive to multipath than the code phase measurement.

Using the notation introduced in Fig. 17, the differential path lengths are $\Delta r_{M,D}^{(p1)} = r_{u,M}^{(p1)} - r_{u,D}^{(p1)}$ and $\Delta r_{M,D}^{(p2)} = r_{u,M}^{(p2)} - r_{u,D}^{(p2)}$. This difference will control how the direct and reflected rays combine, and the resulting carrier phase error is shown in Fig. 21. Using the usual relationships for an oblique triangle, the phase error can be written as

$$\begin{aligned} \nu_{\phi}^{(p1)} &= \frac{\pi \text{ mod } \Delta r_{M,D}}{\lambda} \\ &\quad + \tan^{-1} \left[\left(\frac{\hat{m}-1}{\hat{m}+1} \right) \tan \left(\frac{\pi \Delta r_{M,D}}{\lambda} \right) \right] \text{ rad} \\ \Delta \nu_{\phi} &= \frac{\lambda}{2\pi} (\nu_{\phi}^{(p2)} - \nu_{\phi}^{(p1)}) \text{ m} \\ V_{\text{NSE}} &= \frac{\Delta \nu_{\phi}}{\partial \Delta \phi} \text{ m} \end{aligned} \quad (21)$$

where the middle equation is the error in the differential phase measurement. This error is generally a few centimeters at the most and very small compared to the corresponding code phase measurement error. In fact, the final equation is the vertical position error due to multipath and is plotted in Fig. 20 for a few reflection depths. As shown, this bias is smaller than the error due to noise for all aircraft heights of interest.

3) *Covariance Analysis:* The vertical accuracy provided when in-track APL's are used in concert with carrier-smoothed code is shown in Figs. 2 and 3. These figures have been introduced previously and show the vertical accuracy versus time for an aircraft approaching San Francisco International Airport. Fig. 2 is for 24 healthy GPS satellites and Fig. 3 assumes that 21 satellites are operational. Both figures are for an aircraft at an altitude of 30 m.

The figures use (4) to estimate the vertical accuracy with the following provisions. For in-track APL's, the G matrix is augmented by the row defined in (19) and (20) for the differential carrier phase measurement. Using the conventions defined in Fig. 16, the first APL has $D = 0$ and is located next to the aircraft touchdown point. The second APL has $D = 1200$ m and is located well down the runway from the touchdown point.

Once again, the satellite measurement errors are assumed to be independent from satellite to satellite. Each error includes noise, multipath and tropospheric modeling error.

The variance of the error in a corrected satellite measurement is given by

$$P_{\Delta\tau,k,k} = 2 \left[\frac{\sigma_{\tau}^2 [C(\ell^{(k)})]}{(1-2\eta)} + \frac{(0.1)^2}{3} \right] + \sigma_{\text{tropo}}^2 (\ell^{(k)}) \quad (22)$$

where $P_{\Delta\tau,k,k}$ is still the element in the k th row and k th column of the covariance matrix. As before, the noise variances ($\sigma_{\Delta\tau}^2(C)$) are given by (14), where $C(\ell^{(k)})$ is the peak power of the k th satellite. However, there are now two pseudolite signals, and so the satellite signals are now blanked for 20% of the time ($1-2\eta = 0.8$). As before, the error due to satellite multipath is assumed to have a variance of $(0.1)^2/3$.

The error variance for the hyperbolic APL measurements is given by

$$\begin{aligned} P_{\Delta\phi,K+1,K+1} &= \frac{2\sigma_{\phi}^2 [C^{(\text{APL})}]}{\eta} + (0.022)^2 \\ &\quad + \sigma_{\text{tropo,APL}}^2(D) \text{ m}^2 \\ \sigma_{\phi}^2 [C^{(\text{APL})}] &= \frac{\left(\frac{\lambda}{2\pi} \right)^2 BW_{C,1\text{-sided}}}{\frac{C^{(\text{APL})}}{N_0 + I_0}} \\ &\quad \cdot \left[1 + \frac{1}{2 \left(\frac{C^{(\text{APL})}}{N_0 + I_0} \right) T_{50}} \right] \text{ m}^2 \end{aligned} \quad (23)$$

where the latter formula is for the noise variance suffered by a Costas loop for tracking carrier phase. This analysis assumes that the wavelength ambiguities have been successfully resolved, but the duty cycle of each APL is assumed to be 10% ($\eta = 0.1$) and the integration gain of the loop is reduced accordingly. The APL signal power ($C^{(\text{APL})}$) is assumed to be equal to the satellite power when the aircraft is 10 km from the airport and grows as the aircraft approaches in accord with the $1/R^2$ propagation law. The loop bandwidth is assumed to be 10 Hz. Finally, the carrier phase multipath suffered by each APL is assumed to add an error with a standard deviation of 2.2 cm.

The tropospheric error model is from (17) for ranging pseudolites, but the range is now only the distance between the two APL's (D). After all, the error which accumulates over the common propagation path will cancel.

As shown in Figs. 2 and 3, in-track APL's do much to mitigate the outages suffered by the GPS constellation operating without augmentation. Importantly, they cause the vertical accuracy to be more nearly independent of time. Recent flight trials seem to substantiate the value of hyperbolic APL's [35], [36].

D. APL Siting Constraints

APL siting is governed by multiple constraints. First, APL's must be sited so they meaningfully augment the user geometry. The line of sight vectors they add to the G

matrix must be those which mitigate the outages of concern at any given airport. At the same time, the APL signals must have an unobstructed path to integrity monitors on the ground.

APL's must also be sited to maximize the probability that their signals reach antennas on the top of the aircraft. Indeed, antennas on top of the aircraft are used to receive the GPS satellite signals, and airframe manufacturers would rather use these top antennas for the APL signals than antennas on the aircraft nose or chin. For Category III, any antenna must be duplicated or triplicated for redundancy, and the cost of a new pair of chin antennas is high—especially since the new antennas must coexist with the crowd of antennas already located near the chin. Additionally, the lever arm, which is the vector connecting the chin antennas to the top antennas, must be estimated in real time. Otherwise, the APL data will not provide a meaningful augmentation to the navigation solution. Flight trials are currently addressing the viability of using the top antennas for the APL signals.

Hyperbolic APL's may be more difficult to site than ranging APL's. First, they require at least two APL's on the airport property. Second, the in-track configuration, used as an example in Section IV-C, places the APL's along the runway and therefore more nearly under the nose of an approaching aircraft. Hence aircraft blockage may be more problematic. However, hyperbolic lines of position can be generated from a variety of APL configurations, and they provide vertical position information which is less dependent on the geometry of the GPS satellites.

V. SUMMARY AND CURRENT VALIDATION WORK

This paper begins with a brief sketch of GPS use for aviation. Specifically, the introduction discusses the use of SPS for en-route flight, terminal area operations, and nonprecision approach. Supplemental use in these flight modes is possible when RAIM is used to detect any faults. Then, the paper introduces SBAS's like the FAA's Wide Area Augmentation System. Such systems increase the accuracy, integrity, continuity, and availability of satellite navigation. They are even capable of supporting Category I precision approach operations with decision heights of 200–350 ft.

However, this paper focuses on the use of LAAS's for all categories of precision approach operations. The introduction describes the overall LAAS architecture and establishes the connection between accuracy, integrity, and continuity.

The body of the paper describes the basic LAAS measurements and accuracy. As discussed, code phase measurements cannot by themselves provide the required accuracy because of noise and multipath. Fortunately, the carrier measurements provide very accurate estimates of the change in aircraft position, and so they can be used to greatly extend the averaging time of the position estimator. With carrier aiding and careful antenna design, code phase measurements do provide the required accuracy. However,

the position quality is still sensitive to the number of GPS satellites on orbit. If LAAS is to become operational, then the GPS signals must be augmented by another source of position information.

APL's can provide additional ranging signals and mitigate outages caused by satellite failures. Single APL's can broadcast GPS-like signals which would be treated much like the satellite signals by the receiver. Two or more APL's can be synchronized to provide hyperbolic lines of position based on measurements of differential carrier phase.

As described in Section IV-D, substantive technical issues remain for pseudolites: the signal design must be completed and siting issues must be resolved. If these challenges are met, then APL's will provide robustness against GPS replenishment slow-downs and uncertainties, GPS satellite failures (especially if they occur in unforeseen bursts), interference, and all other performance vagaries.

In general, LAAS requires more definition of the exact integrity algorithm, where this work must define the division of responsibility between the ground equipment and the avionics. The efficacy of the proposed VHF data link must also be explored. This study must establish that adequate VHF capacity exists within the aeronautical navigation bands for multiple LAAS installations at nearby airports. Finally, the LAAS data format must be defined.

APPENDIX A: ACCURACY REQUIRED FOR INTEGRITY AND CONTINUITY

This appendix presents a thought experiment to quantify the relationship between accuracy, integrity, and continuity. This experiment is based on the integrity monitoring algorithm shown in Fig. 22. The algorithm shown in the figure is certainly simplified relative to reality, but it does capture enough of the problem to permit a meaningful analysis.

As shown in Fig. 22, K satellites are observed and processed by two independent chains. Each chain includes a ground reference receiver which makes measurements and passes data to a receiver in the aircraft. The airborne receiver makes its own measurements and applies the differential corrections. After airborne processing is complete, the nominal pseudorange errors for chains #1 and #2 are $\Delta\mathcal{L}_1 = [\Delta\mathcal{L}_1^{(k1)} \dots \Delta\mathcal{L}_1^{(K)}]^T$ and $\Delta\mathcal{L}_2 = [\Delta\mathcal{L}_2^{(k1)} \dots \Delta\mathcal{L}_2^{(K)}]^T$, respectively. These pseudorange errors are assumed to be Gaussian with zero mean and result in position errors given by

$$\begin{aligned}\varepsilon_{z1} &= A_{u,v}\Delta\mathcal{L}_1 + b_v \\ \varepsilon_{z2} &= A_{u,v}\Delta\mathcal{L}_2\end{aligned}$$

where $A_{u,v}$ is the third row of the matrix given by (3). As indicated in Fig. 24, processing chain #1 suffers from some sort of fault beyond the effects of noise. This fault manifests itself as a bias in its vertical position estimate (b_v). This fault would appear at the output of chain #2 and our analysis could be repeated with no meaningful change to the final outcomes presented here. However, a common mode fault would present a much more difficult situation.

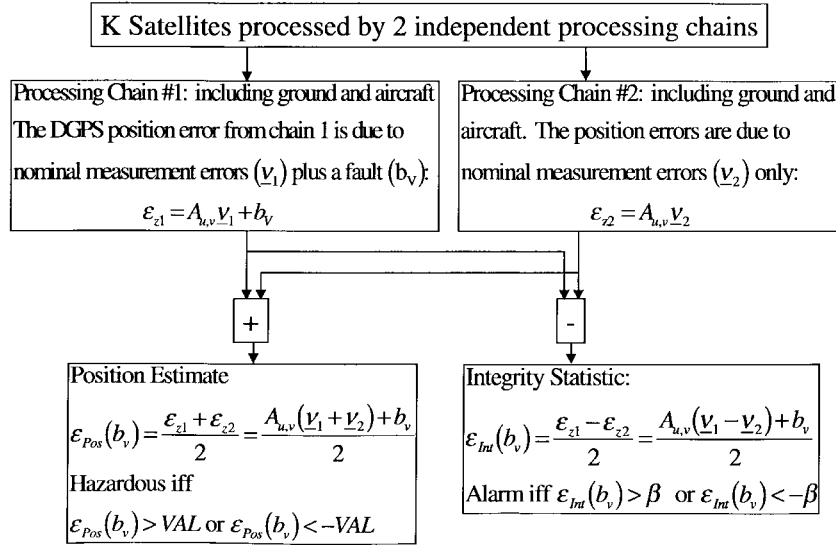


Fig. 22. LAAS signal processing model for computing a relationship between accuracy, integrity, and continuity.

The moments of the position error are

$$E\{\varepsilon_{z1}\} = b_v \quad E\{\varepsilon_{z2}\} = 0$$

$$\sigma_{z1}^2 = \sigma_{z2}^2 = \sigma_z^2 = A_{u,v} P_v A_{u,v}^T = (G^T P_v^{-1} G)_{3,3}^{-1}$$

The landing system shown in Fig. 22 estimates its vertical position by simply averaging and the resulting position error is given by

$$\varepsilon_{Pos}(b_v) = \frac{\varepsilon_{z1} + \varepsilon_{z2}}{2} = \frac{A_{u,v}(\underline{v}_1 + \underline{v}_2) + b_v}{2}.$$

If the absolute value of this total error is greater than a specified VAL and the pilot is not warned, then an integrity failure has occurred.

To prevent such a failure, the integrity processor computes the residual between either of the individual position estimates and the average. This integrity statistic is given by

$$\varepsilon_{Int}(b_v) = \frac{\varepsilon_{z1} - \varepsilon_{z2}}{2} = \frac{A_{u,v}(\underline{v}_1 - \underline{v}_2) + b_v}{2}.$$

Fortunately, the bias error appears here also. The integrity processor sends an alarm if the absolute value of the integrity statistic exceeds a threshold β .

Continuity requires that the total alarm rate be manageably small under nominal conditions, and so we write

$$\Pr(\text{alarm}|b_v = 0) = \Pr(\varepsilon_{Int}(0) > \beta \cup \varepsilon_{Int}(0) < -\beta)$$

$$= 2 \operatorname{erfc}_* \left(\frac{\beta}{\sigma_z / \sqrt{2}} \right) \Rightarrow \beta = k_{fa} \frac{\sigma_z}{\sqrt{2}}$$

where

$$\operatorname{erfc}_*(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{z^2}{2} \right] dz$$

$$\approx \frac{1}{2} \exp \left[-\frac{x^2}{2} \right] \quad x \geq 0$$

$$E\{\varepsilon_{Int}^2(b_v = 0)\} = \frac{\sigma_z^2}{2}. \quad (\text{A.1})$$

If we use the approximation in (A.1), then

$$\beta = \sqrt{-2 \ln \Pr\{\text{alarm}|b_v = 0\}} \frac{\sigma_2}{\sqrt{2}}.$$

As mentioned earlier, integrity and continuity are competing requirements. Integrity demands that the probability of hazardously misleading information (HMI) be between approximately 10^{-10} (Category III) and 10^{-8} (Category I)

$$\Pr(\text{HMI}) = \Pr[\text{large error and no alarm}]$$

$$= \Pr[(\varepsilon_{Pos}(b_v) > VAL \cup \varepsilon_{Pos}(b_v) < -VAL)$$

$$\cap (-\beta < \varepsilon_{Int}(b_v) < \beta)]$$

$$= \Pr \left[\left(\varepsilon_{Pos}(0) > VAL - \frac{b_v}{2} \right. \right.$$

$$\cup \left. \varepsilon_{Pos}(0) < -VAL - \frac{b_v}{2} \right)$$

$$\cap \left(-\beta - \frac{b_v}{2} < \varepsilon_{Int}(0) < \beta - \frac{b_v}{2} \right) \Big].$$

This calculation will require the marginal and joint moments of $\varepsilon_{Int}(0)$ and $\varepsilon_{Pos}(0)$

$$E\{\varepsilon_{Pos}^2(0)\} = E\{\varepsilon_{Int}^2(0)\} = \frac{\sigma_z^2}{2}$$

$$E\{\varepsilon_{Pos}(0)\varepsilon_{Int}(0)\} = 0.$$

$\varepsilon_{Int}(0)$ and $\varepsilon_{Pos}(0)$ have equal variance, because we assume the errors made by processing chain #1 are not correlated with the errors made by processing chain #2. $\varepsilon_{Int}(0)$ and $\varepsilon_{Pos}(0)$ are not correlated because $\sigma_{z1}^2 = \sigma_{z2}^2 =$

σ_z^2 . This lack of correlation allows

$$\begin{aligned} \Pr(\text{HMI}) &= \left[\Pr\left(\varepsilon_{\text{Pos}}(0) > VAL - \frac{b_v}{2}\right) \right. \\ &\quad \left. + \Pr\left(\varepsilon_{\text{Pos}}(0) < -VAL - \frac{b_v}{2}\right) \right] \\ &\quad \cdot \Pr\left(-\beta - \frac{b_v}{2} < \varepsilon_{\text{Int}}(0) < \beta - \frac{b_v}{2}\right) \\ &= \left[\text{erfc}_*\left(\frac{VAL - b_v/2}{\sigma_z/\sqrt{2}}\right) \right. \\ &\quad \left. + \text{erfc}_*\left(\frac{VAL + b_v/2}{\sigma_z/\sqrt{2}}\right) \right] \\ &\quad \cdot \left[\text{erfc}_*\left(\frac{b_v/2 - \beta}{\sigma_z/\sqrt{2}}\right) - \text{erfc}_*\left(\frac{\beta + b_v/2}{\sigma_z/\sqrt{2}}\right) \right]. \end{aligned}$$

The precision approach of aircraft requires that $\Pr(\text{HMI})$ be small regardless of the value of b_v , subject to the constraint that $\Pr(\text{alarm}|b_v = 0)$ is also acceptably small. The value of b_v which maximizes the $\Pr(\text{HMI})$ can be approximated in closed form as follows. For reasonable values of β , VAL , and b_v we can write

$$\begin{aligned} \Pr(\text{HMI}) &\approx \Pr(\varepsilon_{\text{Pos}}(0) > VAL - b_v/2) \\ &\quad \cdot \Pr(\varepsilon_{\text{Int}}(0) > -\beta + b_v/2) \\ &= \text{erfc}_*\left(\frac{VAL - b_v/2}{\sigma_z/\sqrt{2}}\right) \text{erfc}_*\left(\frac{b_v/2 - \beta}{\sigma_z/\sqrt{2}}\right) \\ &\approx \frac{1}{4} \exp\left[\frac{-VAL^2 + b_v VAL - b_v^2/4}{\sigma_z^2} \right. \\ &\quad \left. + \frac{-\beta^2 + b_v \beta - b_v^2/4}{\sigma_z^2} \right] \quad (\text{A.2}) \end{aligned}$$

where the last line has made use of the approximation in (A.1).

The bias which maximizes this probability will also maximize $\ln 4 \Pr(\text{HMI})$, and so

$$\begin{aligned} \frac{\partial}{\partial b_v} \left[\frac{-VAL^2 + b_v VAL - b_v^2/4}{\sigma_z^2} \right. \\ \left. + \frac{-\beta^2 + b_v \beta - b_v^2/4}{\sigma_z^2} \right] \Big|_{b_{v,\max}} = 0 \end{aligned}$$

which yields

$$b_{v,\max} = VAL + \beta = VAL + k_{fa}\sigma.$$

Finally, vertical protection level (VPL) is defined as the value of VAL which can be protected with a specified $\Pr(\text{HMI})$. This value can be computed from (A.2) as

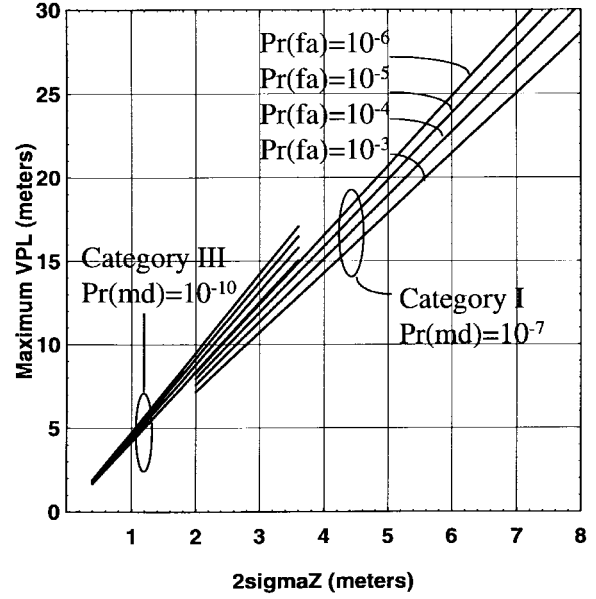


Fig. 23. Maximum VPL versus the vertical accuracy (2 sigma) of one of the two channels. Probability of missed detection and probability of false alarm are the parameters.

follows:

$$\begin{aligned} \ln[\Pr(\text{HMI})] - \ln\left[\frac{1}{4}\right] &= \frac{-VPL^2 + b_{v,\max} VPL - b_{v,\max}^2/4}{\sigma_z^2} \\ &\quad + \frac{-\beta^2 + b_{v,\max} \beta - b_{v,\max}^2/4}{\sigma_z^2} \\ VPL &= \left[k_{fa} + \sqrt{4 \left[\ln\left(\frac{1}{4}\right) - \ln \Pr(\text{HMI}) \right]} \right] \frac{\sigma_z}{\sqrt{2}} \\ &= \left[\sqrt{-2 \ln \Pr(\text{alarm}|b_v = 0)} \right. \\ &\quad \left. + \sqrt{4 \left[\ln\left(\frac{1}{4}\right) - \ln \Pr(\text{HMI}) \right]} \right] \frac{\sigma_z}{\sqrt{2}}. \end{aligned}$$

This function is plotted in Fig. 23 as a function of $2\sigma_z$, with $\Pr(\text{HMI})$ and $\Pr(\text{alarm}|b_v = 0)$ as parameters. As shown, it is directly proportional to $2\sigma_z$ and only varies slowly with either $\Pr(\text{HMI})$ or $\Pr(\text{alarm}|b_v = 0)$. This closed form approximation for VPL is consistent with the more exact analysis presented in [20].

If $\Pr(\text{HMI}) = 10^{-10}$ and $\Pr(\text{alarm}|b_v = 0) = 10^{-5}$, then

$$VPL \approx 10\sigma_z = 10\sqrt{(G^T P^{-1} G)_{3,3}^{-1}}.$$

These values were used in Table 1.

APPENDIX B: UPPER BOUND ON MULTIPATH DOPPLER AT THE REFERENCE RECEIVER

This brief appendix derives an approximation for multipath Doppler shift, when the reflection occurs below the reference antenna as shown in Fig. 7. Recall that the Doppler shift is the time rate of change of the differential

path length, when that path length is measured in radians. In other words

$$\begin{aligned}\dot{\theta}_{MD} &= \frac{2\pi}{\lambda} 2h_m \cos \epsilon l \frac{d\epsilon l}{dt} \text{ rad/s} \\ &= \frac{2\pi}{\lambda} 2h_m \cos \epsilon l \frac{d\epsilon l}{d\gamma} \frac{d\gamma}{dt} \text{ rad/s} \quad (\text{B.1})\end{aligned}$$

where $d\epsilon l/dt$ is the time rate of change of the elevation angle of the satellite. In addition, (B.1) shows the relationship between the elevation angle (ϵl) and γ , where

$$\frac{d\gamma}{dt} = \frac{4\pi}{24(60)60} \text{ rad/s}$$

since any GPS satellite passes overhead twice per day.

From the usual relations for oblique triangles, we have

$$\begin{aligned}\epsilon l &= \tan^{-1} \left[k_R \tan \left(\frac{\pi - \gamma}{2} \right) \right] - \frac{\gamma}{2} \\ \frac{d\epsilon l}{d\gamma} &\leq \frac{-1}{2} \left[1 + \frac{k_R}{\cos^2 \left(\frac{\pi - \gamma}{2} \right) + k_R^2 \sin^2 \left(\frac{\pi - \gamma}{2} \right)} \right] \\ k_R &= \frac{R_{GPS} - R_E}{R_{GPS} + R_E}.\end{aligned}$$

This result is an upper bound, because it assumes that the satellite rises perpendicular to the horizon. If it does not, the relative Doppler shift will be smaller. This equation is combined with (B.1) to provide the curves in Fig. 12.

APPENDIX C: MULTIPATH EFFECT ON CODE PHASE MEASUREMENTS

In this brief appendix, we will bound the size of the code phase errors due to multipath. The derivation in this appendix is intended to be straightforward and instructive. However, it is not as complete as the more sophisticated treatments of code phase multipath found in [28] or [40]. These deeper analyses include the nonlinearities used in most delay lock loops and the effects of finite precorrelation bandwidth. Neither of these effects is included here, but neither is required to understand and approximately quantify the fundamental effect of multipath.

Figs. 24 and 25 show the effect of multipath on the shape of the correlation peak. Fig. 24 is for constructive interference and so the composite (direct plus reflected) correlation function is the sum of the two individual correlation functions. In other words, this figure assumes that the relative phase between the direct and reflected rays is zero. In contrast, Fig. 25 assumes that the relative phase is $\pi/2$, and so the composite function is the direct minus the reflected.

We assume that the peak of the direct correlation function is normalized to unity and so the peak of the correlation function for the reflected ray is equal to the magnitude of the reflected ray normalized by the magnitude of the direct ray ($\hat{m}$). In addition, the width of the C/A code chip is now measured in meters ($T_C = 300$ m). The direct correlation function rises from $(0, -T_C)$ to $(1, 0)$ and then falls from $(1,$

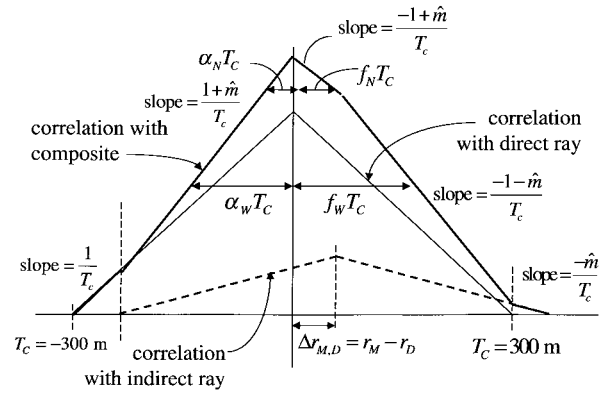


Fig. 24. Correlation peak distorted by constructive multipath interference.

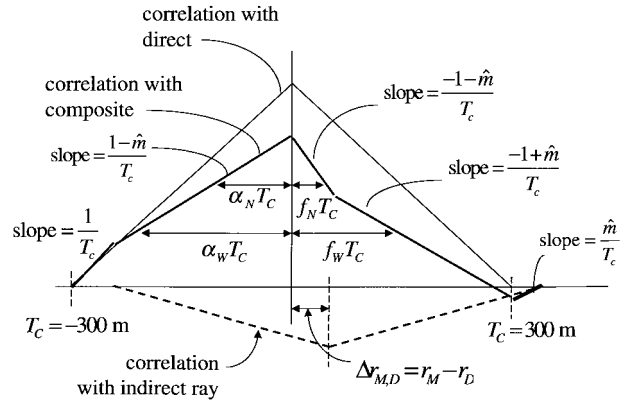


Fig. 25. Correlation peak distorted by destructive multipath interference.

0) to $(0, T_C)$. In addition, the path length for the reflected ray is $\Delta r_{M,D} = r_M - r_D$ meters longer than the path length for the direct ray. Consequently, the peak of the correlation function for the reflected ray occurs at $\Delta r_{M,D}$ meters.

With these definitions, we can label the slopes of all the line segments shown in Figs. 24 and 25. Starting from the left on Fig. 24, the composite function has slope $1/T_C$ because the reflected ray has not yet arrived. After it arrives, the composite slope becomes $(1 + \hat{m})/T_C$ because the two rays are in phase. Immediately after the peak, the slope becomes $(-1 + \hat{m})/T_C$ because the direct correlation function is now falling, but the reflected function is still rising. After $\Delta r_{M,D}$, both functions are falling and so the slope is $(-1 - \hat{m})/T_C$. Finally, only the reflected function contributes to the composite and so the slope is $-\hat{m}/T_C$. Similar values appear on Fig. 25.

Now, we turn our attention to finding the pseudorange error due to multipath. Consider the sample times shown on both figures. In both cases, samples on the rising edge occur either $\alpha_N T_C$ or $\alpha_W T_C$ seconds before time zero, and the samples on the falling edge occur either $f_N T_C$ or $f_W T_C$ past time zero. The samples $\alpha_N T_C$ and $f_N T_C$ are used by a correlator with narrow spacing, and so $d_N = \alpha_N + f_N$ is the correlator spacing. Similarly, the wide correlator uses $d_W = \alpha_W + f_W$.

Table 2

Code Phase Multipath Error Bounds

	Short delayed reflection $\Delta r_{M,D} < fT_C$	long delayed reflection $\Delta r_{M,D} \geq fT_C$
constructive interference	$\Delta r_{M,D} \frac{\hat{m}}{1+\hat{m}}$	$\frac{dT_C \hat{m}}{2}$
destructive interference	$-\Delta r_{M,D} \frac{\hat{m}}{1-\hat{m}}$	$\frac{-dT_C \hat{m}}{2}$

In either event, we can write

$$\begin{aligned} \alpha + f &= d \\ \text{error} &= (f - \alpha)T_C/2 \end{aligned} \quad (\text{C.1})$$

where α , f , and d are for either a wide or narrow correlator. If the correlator spacing is such that $fT_C \leq \Delta r_{M,D}$, then we can write the last needed equation by balancing the value sampled on the leading edge with the value sampled on the falling edge

$$-\left(\frac{1+\hat{m}}{T_C}\right)\alpha T_C = \left(\frac{-1+\hat{m}}{T_C}\right)fT_C$$

where this equation is for constructive interference (i.e., Fig. 24). For $fT_C \leq \Delta r_{M,D}$ and destructive interference, the equation which nulls the input to the delay lock loop is

$$-\left(\frac{1-\hat{m}}{T_C}\right)\alpha T_C = -\left(\frac{1+\hat{m}}{T_C}\right)fT_C.$$

If $fT_C > \Delta r_{M,D}$ and the interference is constructive, then we have

$$\begin{aligned} -\left(\frac{1+\hat{m}}{T_C}\right)\alpha T_C &= \left(\frac{-1+\hat{m}}{T_C}\right)\Delta r_{M,D} - \left(\frac{1+\hat{m}}{T_C}\right) \\ &\cdot (fT_C - \Delta r_{M,D}). \end{aligned}$$

Finally if $fT_C > \Delta r_{M,D}$ and the interference is destructive, then

$$\begin{aligned} -\left(\frac{1-\hat{m}}{T_C}\right)\alpha T_C &= -\left(\frac{1+\hat{m}}{T_C}\right)\Delta r_{M,D} - \left(\frac{1-\hat{m}}{T_C}\right) \\ &\cdot (fT_C - \Delta r_{M,D}). \end{aligned}$$

These last four equations taken together with (C.1) can be solved for the error as shown in Table 2. The data in this table are plotted in Fig. 13.

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Per Enge (Member, IEEE), for a photograph and biography, see this issue, p. 15.